

A NON-ABELIAN DISCRETE STOKES LAW FOR S_N -VALUED CONNECTIONS ON 2-COMPLEXES

J. FUENTES AGUILAR^{ID}

ABSTRACT. Let S_N denote the symmetric group on a set of N elements. We attach to each vertex of a 2-cell complex an empirical measure on N states, couple neighbouring fibres by an entropic optimal transport problem with regularisation ε , and study the discrete connection that results through the holonomy of its loop transports. As ε tends to zero each edge transport concentrates on a permutation matrix, and the loop transport $\mathcal{H}_\gamma$ at a basepoint x_0 defines a representation $\rho : \pi_1(G, x_0)^{\text{op}} \rightarrow S_N$ of the opposite of the fundamental group of the 1-skeleton. The character spectrum $p_k(\gamma) = N^{-1} \text{Tr } \mathcal{H}_\gamma^k$ and the abelianisation $\eta(\gamma) = \text{sign det } \mathcal{H}_\gamma \in \{\pm 1\}$ are the conjugacy invariants of $\mathcal{H}_\gamma$. A non-abelian discrete Stokes law gives the boundary holonomy of a simply connected region D as the tree-ordered product $\mathcal{H}_{\partial D} = \prod_c g_c^{-1} \mathcal{H}_c g_c$. Its abelianisation yields the sum rule $\prod_{c \in F(\Sigma)} \eta(c) = +1$ on every closed oriented 2-surface Σ . On the torus the abelianisation produces a $\mathbb{Z}_2 \times \mathbb{Z}_2$ topological signature $(\eta_\alpha, \eta_\beta)$ that is invariant under all basepoint gauge transformations and invisible to every contractible observable. The signature is a gauge-invariant class in the first $\mathbb{Z}_2$ -cohomology $H^1(T^2, \mathbb{Z}_2)$ of the torus, and its four values exhaust that group. Newton's identities provide a finite character-spectrum test of conservativeness on the $r = |E| - |V| + 1$ generators of $\pi_1(G, x_0)$. The $\varepsilon \rightarrow 0$ limit concentrates exponentially fast when the optimal assignment is uniformly gapped and only polynomially when it is degenerate, and at positive ε a doubly stochastic Stokes deficit valued in squared face determinants measures the failure of conservativeness. Explicit permutation connections realise all four sectors of the signature and confirm the closed-surface sum rule.

Corresponding author: J. Fuentes Aguilar, jesus.fuentes@uni.lu, University of Luxembourg, Belval, Luxembourg.

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1. INTRODUCTION

Let $\mathcal{X}$ be a smooth base manifold and $F : \mathcal{X} \times \Theta \rightarrow \mathcal{Y}$ a smooth map with parameter domain Θ , equipped with a probability measure μ_Θ on Θ . An ensemble of N parameter vectors $\{\theta_i\}_{i=1}^N$ drawn independently from μ_Θ defines, at each $x \in \mathcal{X}$, the empirical fibre

$$\hat{\mu}_x = \frac{1}{N} \sum_{i=1}^N \delta_{j^1 f_{\theta_i}(x)}, \quad j^1 f_\theta(x) = (f_\theta(x), df_\theta(x)), \quad (1)$$

of jets of order one valued in the jet space $\mathcal{Z} = \mathcal{Y} \times \mathbb{R}^{k \times n}$. The data of $\hat{\mu}_x$ is invariant under permutation of the ensemble indices and independent of any labelling assigned to the sections $\{f_{\theta_i}\}$. The family $\{\hat{\mu}_x\}_{x \in \mathcal{X}}$ is the empirical content of an S_N -bundle over $\mathcal{X}$, with S_N acting on the ensemble indices by permutation.

An agnostic correspondence of labels between fibres at neighbouring inputs x_α, x_β is supplied by an entropic optimal transport (OT) problem with the Gibbs kernel

$$K_{\alpha\beta,ij}^{(\varepsilon)} = \exp(-C_{\alpha\beta,ij}/\varepsilon), \quad C_{\alpha\beta,ij} = d_{\mathcal{Z}}^2(z_i(x_\alpha), z_j(x_\beta)), \quad (2)$$

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Corresponding author: J. Fuentes Aguilar, jesus.fuentes@uni.lu.

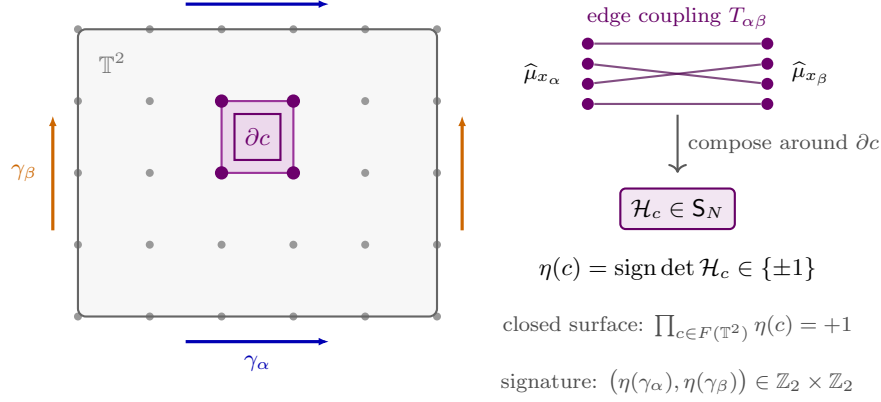


FIGURE 1. The base manifold is the torus $\mathbb{T}^2$ (left), presented with the non-contractible generators $\gamma_\alpha, \gamma_\beta$ as seam identifications, and a single plaquette c is highlighted. At every sampled input x , an ensemble of N sections drawn from μ_Θ induces an empirical jet fibre $\hat{\mu}_x$ (right, top). Between neighbouring inputs we form an entropic OT coupling $T_{\alpha\beta}$, whose $\varepsilon \downarrow 0$ limit under a uniform assignment gap is permutation-valued. Composing edge couplings around the plaquette boundary ∂c yields the holonomy $\mathcal{H}_c \in \mathbf{S}_N$, with cycle type given by the character spectrum $\{p_k(c)\}$ of Definition 15 and sign given by the abelianisation $\eta(c)$ of Definition 16. The non-abelian discrete Stokes law of Theorem 25 assembles face holonomies into boundary holonomies. On a closed oriented surface the law delivers the sum rule $\prod_{c \in F(\Sigma)} \eta(c) = +1$ of Corollary 27, and on the torus the $\mathbb{Z}_2 \times \mathbb{Z}_2$ topological signature $(\eta(\gamma_\alpha), \eta(\gamma_\beta))$ of Corollary 28, which is the central empirical object of the paper.

in which $d_{\mathcal{Z}}$ balances values and Jacobians under a Levi–Civita alignment of tangent frames. The unique entropic coupling $\Pi_{\alpha\beta}^{(\varepsilon)}$ with uniform marginals rescales to a doubly stochastic edge map $T_{\alpha\beta} = N \Pi_{\alpha\beta}^{(\varepsilon)}$ on the neighbourhood graph $G = (V, E)$ over $\mathcal{X}$, and the family $\{T_{\alpha\beta}\}$ defines a discrete connection on the empirical bundle. Sending $\varepsilon \downarrow 0$ under a uniform suboptimality gap concentrates $T_{\alpha\beta}$ on a permutation matrix in $\mathbf{S}_N$, while exact linear assignment at $\varepsilon = 0$ produces $T_{\alpha\beta} \in \mathbf{S}_N$ directly. We refer to these three settings as (A) entropic, (B) assignment limit, and (C) exact assignment, and develop them in parallel below.

We say the connection $\{T_{\alpha\beta}\}$ is conservative when the loop product

$$\mathcal{H}_\gamma = T_{x_{L-1} \rightarrow x_L} \cdots T_{x_1 \rightarrow x_2} T_{x_0 \rightarrow x_1}, \quad \gamma = (x_0, x_1, \dots, x_L = x_0), \quad (3)$$

returns to the identity along every closed path γ based at $x_0 \in V$. In regimes (B) and (C) the loop transport takes values in the symmetric group, and the assignment $\gamma \mapsto \mathcal{H}_\gamma$ descends to a representation

$$\rho : \pi_1(G, x_0)^{\text{op}} \longrightarrow \mathbf{S}_N, \quad (4)$$

of the opposite of the fundamental group of G at x_0 in $\mathbf{S}_N$, gauge-equivariant under basepoint relabelling by conjugation. Triviality of ρ on every cycle is equivalent to the existence of a global labelling of ensemble members across $\mathcal{X}$, and a non-trivial ρ records the obstruction to such a labelling. Figure 1 summarises the construction.

The conjugacy invariants of $\mathcal{H}_\gamma$ are the natural diagnostics of the connection. The character spectrum

$$p_k(\gamma) = \frac{1}{N} \text{Tr } \mathcal{H}_\gamma^k, \quad k = 1, \dots, N-1, \quad (5)$$

recovers the cycle structure of $\mathcal{H}_\gamma$ via Newton's identities, and the abelianisation

$$\eta(\gamma) = \text{sign det } \mathcal{H}_\gamma \in \{\pm 1\} \quad (6)$$

factors ρ through $\rho^{\text{ab}} : \pi_1(G, x_0) \rightarrow \mathbb{Z}_2$. The Frobenius scalar that has played the role of holonomy proxy in earlier diagnostic work [29, 47] appears as a contraction of the first character, $h_{\text{op}}(\gamma) = \sqrt{2(1 - \text{Re } p_1(\gamma))}/N$.

The construction depends only on the data of an empirical measure on a finite set at each vertex of G together with a cost between neighbouring fibres, and the source of that data is left open. Configurations of an N -state lattice model, crystal grain orientations valued in a finite point group, S_N -coloured spin foams [9], and Bloch frames of band structures over a Brillouin torus all furnish empirical fibres on N states over a 2-cell complex. The closed-surface sum rule of (8) and the torus signature of (9) read these fibres through the $\mathbb{Z}_2$ -cohomology of the complex. The theory is developed in the abstract from §2 onward, and the neural-ensemble instantiation is presented in §A.

1.1. Main Results. Fix a simply connected 2-cell subcomplex $D \subset G$, a basepoint x_0 on ∂D , a spanning tree $\mathcal{T}_D$ of the vertices of D rooted at x_0 , and a tree-compatible enumeration $(c_1, \dots, c_n)$ of the faces $F(D)$. With $g_c = \mathcal{H}_{\pi_c}$ the parallel transport along the unique tree path π_c from x_0 to a chosen corner of c , Theorem 25 establishes the non-abelian discrete Stokes law

$$\mathcal{H}_{\partial D} = \prod_{k=1}^n g_{c_k}^{-1} \mathcal{H}_{c_k} g_{c_k} \quad \text{in } S_N \quad (7)$$

in regime (C), with the product read left-to-right under the opposite-group convention of (4). The right-hand side is independent of the tree, the corners, and the ordering, beyond tree-compatibility.

Applying $\eta = \text{sign det}$ to (7) yields, by multiplicativity and conjugation invariance, the abelian discrete Stokes law $\eta(\partial D) = \prod_{c \in F(D)} \eta(c)$ of Corollary 26. Summing over faces of any closed oriented 2-surface Σ triangulated by G , and accounting for the two oriented traversals of every edge, produces the closed-surface sum rule

$$\prod_{c \in F(\Sigma)} \eta(c) = +1 \quad (8)$$

of Corollary 27. The non-contractible generators of Σ remain undetected by (8), and on the torus T^2 , with non-contractible generators $\gamma_\alpha, \gamma_\beta$ of $\pi_1(T^2, x_0) \cong \mathbb{Z}^2$, the pair

$$(\eta_\alpha, \eta_\beta) := (\eta(\gamma_\alpha), \eta(\gamma_\beta)) \in \mathbb{Z}_2 \times \mathbb{Z}_2 \quad (9)$$

is, by Corollary 28, gauge invariant at the basepoint, invariant under continuous deformation of $\gamma_\alpha, \gamma_\beta$ within their based homotopy classes, and invisible to every diagnostic evaluated on contractible loops or on local summaries of the field. The four sectors $(\eta_\alpha, \eta_\beta) \in \{\pm 1\}^2$ partition the space of regime-(C) connections on T^2 into gauge-invariant topological classes. Proposition 29 realises these classes as the elements of $H^1(T^2, \mathbb{Z}_2)$, exhibited surjectively by the seam-twist construction.

The reach of the signature is delimited on both sides. It detects the class in $H^1(T^2, \mathbb{Z}_2)$ carried by the two non-contractible generators, and the character spectrum refines this to the full cycle type of $\mathcal{H}_\gamma$ on any generator of $\pi_1(G, x_0)$. It does not separate connections that agree on those generators, it assigns the same value to every contractible loop, and it carries no quadratic refinement of the intersection form, so it records a cohomology class and not a finer geometric datum on the torus.

Theorem 17 reduces the conservativeness test to a finite character-spectrum criterion on the $r = |E| - |V| + 1$ generators of $\pi_1(G, x_0)$ through Newton's identities. Lemma 18 resolves the assignment limit into two sub-regimes, with exponential concentration $\|\Pi^{(\varepsilon)} - P_\sigma\|_1 \leq K e^{-\Delta_*/\varepsilon}$ when the linear-assignment minimiser is uniformly gapped (sub-regime (B-gap)), and polynomial

concentration when the minimiser is degenerate (sub-regime (B-deg)). Proposition 31 establishes a doubly stochastic Stokes deficit

$$\prod_{c \in F(\Sigma)} \det \mathcal{H}_c = \prod_{e \in E(\Sigma)} (\det T_e)^2 \in [0, 1] \quad (10)$$

on a closed oriented Σ in regime (A), with right-hand side equal to 1 if and only if every T_e is a permutation, recovering (8) in the $\varepsilon \downarrow 0$ limit of (B-gap).

Explicit connections realise the signature and exhaust its four sectors, a canonical example sitting at $(\eta_\alpha, \eta_\beta) = (-1, -1)$ and seam twists reaching the rest. The abelianisation η carrying the signature is a regime-(C) invariant, and in the entropic regime (A) the transports around long loops mix to the uniform stationary matrix, their determinants underflow, and the signature is not numerically resolvable, so the topological reading rests on the exact-assignment connection. Higher characters $\text{Re } p_k$ for $k \geq 2$ correlate weakly with $\text{Re } p_1$, confirming that h_{op} projects onto the first power trace and discards information the full character spectrum retains. These realisations are carried out in Appendix A and, for higher genus, in Appendix B.

1.2. Related Work. Vector diffusion maps build a connection on a data manifold by transporting tangent information between neighbouring samples and assembling a connection Laplacian that respects parallel transport, recovering manifold structure from heat kernels [46], and parallel transport unfolding adds a metric connection on the same data manifold to improve geodesic distance estimates [15]. The empirical S_N -bundle of this paper carries a connection of a different kind, defined by entropic OT between empirical fibres and independent of the ambient geometry of $\mathcal{X}$, and the object of study is the holonomy of that connection rather than a Laplacian built from it. Discrete connections on principal bundles and the discrete exterior calculus place differential-geometric operators on cell complexes [21, 36], and cellular sheaves with their Laplacians give a cohomological account of data attached to a complex [27]. The connection here is non-abelian and permutation-valued, built from optimal transport rather than from a fixed structure group, and its holonomy is read for the $\mathbb{Z}_2$ -cohomology class it carries rather than through a Laplacian or a sheaf-theoretic spectral sequence.

Parallel transport in Wasserstein space is by now classical, and recent work examines holonomy when additional structure is present [1, 6, 37]. Entropic OT produces stable edge couplings between empirical fibres in the discrete setting, and the composition of these couplings around a loop is the holonomy of the discrete Gibbs connection studied here.

For restricted Boltzmann machines, Magnot [39] defines group-valued holonomies on the bipartite graph and introduces a context index measuring global inconsistency, with fibres carrying discrete group-valued weights rather than empirical jets over an input manifold. The S_N -holonomy here is read instead from optimal transport between empirical measures, and its conjugacy invariants descend to the $\mathbb{Z}_2$ -cohomology of the complex.

The learning-theoretic background sits alongside this construction rather than inside it. Information geometry equips parameter space with a statistical metric exploited by natural-gradient and Riemannian methods [2, 4, 5, 10, 42], optimal transport casts training as a gradient flow in Wasserstein space [31, 43, 51], the loss landscape has been read through Hessian spectra and mode connectivity [22, 23, 25, 26, 30, 33, 40], group and gauge equivariant networks build symmetry into the architecture [14, 17, 18, 35], and alignment and model merging methods produce a single global match between trained networks in parameter or representation space [3, 48, 49, 53, 54]. The present paper takes the parameter measure μ_Θ as given, couples the resulting empirical fibres by entropic OT, and studies the conjugacy invariants of the loop products of that coupling.

1.3. Organisation Of The Paper. Section 2 develops the empirical fibre bundle, introduces jet enrichment, and defines the empirical fibres of (1). Section 3 constructs the holonomy representation (4), the character spectrum (5), the abelianisation (6), the three transport regimes (A)–(C) in

§3.2, and the gap-regime dichotomy of Lemma 18. Section 4 proves the non-abelian discrete Stokes law (7), the closed-surface sum rule (8), the topological signature (9), and the doubly stochastic deficit (10). Section 5 treats the discrete approximation viewpoint and stability under mesh refinement, and Section 6 closes with remarks and outlook. The numerical work is gathered in the appendices, with Appendix A reporting synthetic results on the torus, including character-spectrum measurements and the realisation of all four topological sectors by prescribed twists, Appendix B a permutation connection on a closed surface of higher genus.

2. GENERAL BACKGROUND

A smooth map $F : \mathcal{X} \times \Theta \rightarrow \mathcal{Y}$ together with a probability measure on Θ produces a geometric structure on the base manifold $\mathcal{X}$. Each parameter draw $\theta \in \Theta$ is a section, and for a fixed input x the local behaviour of these sections assembles into a fibre. For a fixed x the fibre depends only on the randomness in θ , and a typical fibre is therefore diffeomorphic to the support of the parameter measure.

We denote by $(\mathcal{X}, g_{\mathcal{X}})$ a smooth n -dimensional Riemannian manifold with metric $g_{\mathcal{X}}$, realised as an embedded submanifold of $\mathbb{R}^m$ with $n \leq m$ and carrying the metric induced from the ambient Euclidean space. The codomain is a subset $\mathcal{Y} \subseteq \mathbb{R}^k$. The smooth map

$$F : \mathcal{X} \times \Theta \rightarrow \mathcal{Y}, \quad F(x, \theta) = f_{\theta}(x),$$

takes $x \in \mathcal{X}$ together with a parameter vector $\theta \in \Theta \subset \mathbb{R}^p$ to an element of $\mathcal{Y}$. A stochastic procedure $\mathcal{A}$ acting on a space $\mathcal{R}$ of seeds and initialisations and equipped with a probability measure $\mu_{\mathcal{R}}$ pushes forward to $\mu_{\Theta} := \mathcal{A}_{\#}\mu_{\mathcal{R}}$, the parameter measure on Θ . We assume that F is smooth in both arguments and that the support of μ_{Θ} is compact.

2.1. The Empirical Fibre Bundle. From the inputs $x \in \mathcal{X}$ and outputs $y \in \mathcal{Y}$ of F , gathering at a fixed x all values produced as θ varies over the support of μ_{Θ} defines the graph of the evaluation map as a bundle over $\mathcal{X}$. Once (x, θ) is specified, y is determined, and y does not serve as an independent coordinate on the fibre.

Definition 1 (Total space and projection). Define

$$E := \{(x, y, \theta) \in \mathcal{X} \times \mathcal{Y} \times \Theta : y = f_{\theta}(x), \theta \in \text{supp}(\mu_{\Theta})\},$$

with projection $\pi : E \rightarrow \mathcal{X}$, $\pi(x, y, \theta) = x$. For $x \in \mathcal{X}$, the fibre is

$$E_x := \pi^{-1}(x) = \{(x, f_{\theta}(x), \theta) : \theta \in \text{supp}(\mu_{\Theta})\}.$$

Varying θ across the support of μ_{Θ} at fixed x sweeps out every output value $f_{\theta}(x)$ together with the parameter that produced it, and y is therefore not an independent coordinate on the fibre.

Lemma 2 (Bundle structure). *The bundle $(E, \pi, \mathcal{X})$ is smoothly isomorphic to the trivial bundle $\mathcal{X} \times \text{supp}(\mu_{\Theta}) \rightarrow \mathcal{X}$. In particular, each fibre E_x is diffeomorphic to $\text{supp}(\mu_{\Theta})$.*

Remark 3 (Meaning of diffeomorphic to $\text{supp}(\mu_{\Theta})$). Under the standing assumptions just stated, all occurrences of diffeomorphic to $\text{supp}(\mu_{\Theta})$ are to be read as diffeomorphic to an open neighbourhood of $\text{supp}(\mu_{\Theta})$ inside the smooth parameter manifold [28]. This avoids pathological cases where the topological support fails to be a manifold, and keeps the fibres in correspondence with the support of μ_{Θ} .

Proof. The map

$$\Phi : \mathcal{X} \times \text{supp}(\mu_{\Theta}) \rightarrow E, \quad (x, \theta) \mapsto (x, f_{\theta}(x), \theta),$$

is a diffeomorphism with inverse $(x, y, \theta) \mapsto (x, \theta)$, and π therefore identifies with projection onto the first factor. $\square$

Definition 4 (Sections of the bundle). Each $\theta \in \text{supp}(\mu_\Theta)$ determines a smooth section

$$s_\theta : \mathcal{X} \rightarrow E, \quad s_\theta(x) = (x, f_\theta(x), \theta),$$

with $\pi \circ s_\theta = \text{id}_\mathcal{X}$. An ensemble is a family of sections sampled from μ_Θ .

2.2. Jet Enrichment. Jets of first order record both the output value and its differential, and they support a metric for comparing local behaviour that respects the differential structure of $\mathcal{X}$ and $\mathcal{Y}$.

Definition 5 (Jet of order one). For $f_\theta : \mathcal{X} \rightarrow \mathcal{Y}$, the jet of order one at x is

$$j^1 f_\theta(x) = (f_\theta(x), \text{d}f_\theta(x)),$$

with $\text{d}f_\theta(x) : T_x \mathcal{X} \rightarrow T_{f_\theta(x)} \mathcal{Y}$ the differential. In local frames this is a $k \times n$ matrix, an element of $\text{Hom}(T_x \mathcal{X}, T_{f_\theta(x)} \mathcal{Y}) \cong \mathbb{R}^{k \times n}$ [45].

Definition 6 (Total space enriched with jets). Define

$$\widehat{E} := \{(x, y, J, \theta) : y = f_\theta(x), J = \text{d}f_\theta(x), \theta \in \text{supp}(\mu_\Theta)\},$$

with projection $\widehat{\pi}(x, y, J, \theta) = x$.

Lemma 7 (Jet bundle structure). $(\widehat{E}, \widehat{\pi}, \mathcal{X})$ is a smooth fibre bundle with typical fibre diffeomorphic to $\text{supp}(\mu_\Theta)$. Equivalently, $\widehat{E}$ is diffeomorphic to $\mathcal{X} \times \text{supp}(\mu_\Theta)$ via $(x, \theta) \mapsto (x, f_\theta(x), \text{d}f_\theta(x), \theta)$.

Proof. Smoothness of $(x, \theta) \mapsto (x, f_\theta(x), \text{d}f_\theta(x), \theta)$ and the same graph argument as above give the claim. $\square$

For fixed x , the pair (y, J) is uniquely determined by θ , and the fibre therefore carries no independent factor of $\mathcal{Y} \times \mathbb{R}^{k \times n}$. The variables y and J are kept in the ambient coordinates because the metric below is defined on them, and the intrinsic fibre remains the support of μ_Θ .

2.3. Empirical Fibres. A finite ensemble passes the theoretical bundle to its empirical counterpart through a sample of parameters from μ_Θ together with the convention of disregarding the labels of ensemble members when comparing fibres.

Definition 8 (Empirical fibre as multiset). Given parameter samples $\{\theta_i\}_{i=1}^N \subset \text{supp}(\mu_\Theta)$, with N the number of sections sampled, the empirical fibre at $x \in \mathcal{X}$ is the multiset

$$\widehat{E}_x := \{(f_{\theta_i}(x), \text{d}f_{\theta_i}(x))\}_{i=1}^N \subset \mathcal{Y} \times \mathbb{R}^{k \times n}.$$

For comparisons on the basis of transport, the associated empirical measure on the jet space $\mathcal{Z} := \mathcal{Y} \times \mathbb{R}^{k \times n}$,

$$\widehat{\mu}_x := \frac{1}{N} \sum_{i=1}^N \delta_{(f_{\theta_i}(x), \text{d}f_{\theta_i}(x))},$$

is the convenient object. Although θ_i appears in the construction, the measure $\widehat{\mu}_x$ carries no memory of those labels in that any permutation $i \mapsto \sigma(i)$ of the ensemble leaves it unchanged. Passing to $\widehat{\mu}_x$ is the step at which the parameter ordering is deliberately shed, since across nearby inputs there is no canonical correspondence of ensemble members and any meaningful comparison must be label-invariant.

A metric on $\mathcal{Z}$ balancing value and jet information, introduced in §3, transports $\widehat{\mu}_x$ to $\widehat{\mu}_{x'}$ between neighbouring inputs and defines a discrete connection [6, 7, 44]. The idealised bundle above serves only to justify that this empirical object is a consistent discretisation of a smooth construction under mesh refinement and increasing ensemble size.

3. HOLONOMY REPRESENTATION AND CHARACTER SPECTRUM

We compare local behaviour across inputs through a metric on jets, transport empirical fibres between neighbours to obtain a discrete connection, and read path dependence around loops by composing these edge transports into a holonomy operator. The central object is the holonomy representation $\rho : \pi_1(G, x_0)^{\text{op}} \rightarrow S_N$, where $G = (V, E)$ is the neighbourhood graph of Definition 10, $x_0 \in V$ a basepoint, S_N the symmetric group on an N -element set, and $^{\text{op}}$ the reversed-composition convention of Definition 14.

The character spectrum $\{p_k(\gamma) = N^{-1} \text{Tr } \mathcal{H}_\gamma^k\}_{k=1}^{N-1}$ (Definitions 14–16) carries the bulk of the structure, and every other invariant in this study, including the scalar Frobenius gap $h_{\text{op}} = \sqrt{2(1 - \text{Re } p_1)/N}$, is a contraction of it. Fibres are compared in a way that is blind to the ordering of models in the ensemble, and any permutation of those indices must therefore leave every resulting quantity unchanged. A quantity counts as an invariant only when it is also unaffected by refinement of the neighbourhood graph and by harmless reparameterisations.

Throughout this section, $\mathcal{Z} := \mathcal{Y} \times \mathbb{R}^{k \times n}$ denotes the jet space, and $\hat{\mu}_x$ the empirical measure on $\mathcal{Z}$ built from an ensemble at x , in the setting of §2.

3.1. A Metric On Jets. To compare jets attached to different features, we must first align their tangent spaces. When $\mathcal{X}$ is embedded in $\mathbb{R}^m$, the ambient coordinates already identify tangent vectors. On a general Riemannian manifold, we use Levi–Civita parallel transport along the chosen edge, written

$$\text{PT}_{x \rightarrow x'} : T_x \mathcal{X} \longrightarrow T_{x'} \mathcal{X}. \quad (11)$$

Definition 9 (Jet metric). Fix non-negative weights $\lambda_{\text{val}}, \lambda_{\text{jac}}$, not both zero. For $z = (y, J)$ attached to x and $z' = (y', J')$ attached to x' , define

$$d_{\mathcal{Z}}^2((x, z), (x', z')) = \lambda_{\text{val}} \|y - y'\|_{\mathbb{R}^k}^2 + \lambda_{\text{jac}} \|J - J' \circ \text{PT}_{x \rightarrow x'}\|_{\text{HS}}^2,$$

where $\|\cdot\|_{\text{HS}}$ denotes the Hilbert–Schmidt norm. The correspondence $T_{f_\theta(x)} \mathcal{Y} \cong \mathbb{R}^k$ ensures Jacobians are always compared in the same Euclidean codomain.

This metric balances agreement in function values and in sensitivities as taking $\lambda_{\text{jac}} = 0$ yields a value (output) comparison, while $\lambda_{\text{val}} = 0$ compares purely the local linearisation. The dependence on the edge (x, x') enters only through the chosen transport.

In practice, for a generic manifold $(\mathcal{X}, g_{\mathcal{X}})$, we take (11) along the short geodesic from x to x' . Provided $\mathcal{X} \subset \mathbb{R}^m$ is specified only as a point cloud with the induced metric, we approximate transport by a simple geometric alignment, i.e. we estimate each tangent space via local principal component analysis, project vectors orthogonally into these spaces, and then align them by a Procrustes fit. Concretely, if $P_x, P_{x'}$ are the orthogonal projectors onto the estimated tangent spaces and $U \Sigma V^\top$ is the singular value decomposition of $P_{x'} P_x$, we transport via $\text{PT}_{x \rightarrow x'} \approx UV^\top$. On a graph of nearest neighbours this alignment is applied edge by edge, and transport along a path is the product of the edge maps.

3.2. Regimes And Standing Assumptions. Three operational regimes differ in the algebraic structure of the edge transports they carry. On each edge the entropic regularisation of the OT problem is governed by a temperature parameter $\varepsilon \geq 0$, with lowering ε intensifying the coupling towards a deterministic matching. The three regimes correspond to keeping ε fixed in (A), sending it to zero in (B), and bypassing the regularisation entirely in (C).

(A) Entropic regime. At a fixed temperature $\varepsilon > 0$, each oriented edge $x_\alpha \rightarrow x_\beta$ receives an entropic OT coupling $\Pi_{\alpha\beta}^{(\varepsilon)}$ between $\hat{\mu}_{x_\alpha}$ and $\hat{\mu}_{x_\beta}$ with uniform marginals, and we set $T_{\alpha\beta} := N \Pi_{\alpha\beta}^{(\varepsilon)}$ as in Definition 10. Each $T_{\alpha\beta}$ is doubly stochastic but need not be invertible. Loop products are therefore well-defined linear maps on probability vectors, though they do not in general carry a group structure.

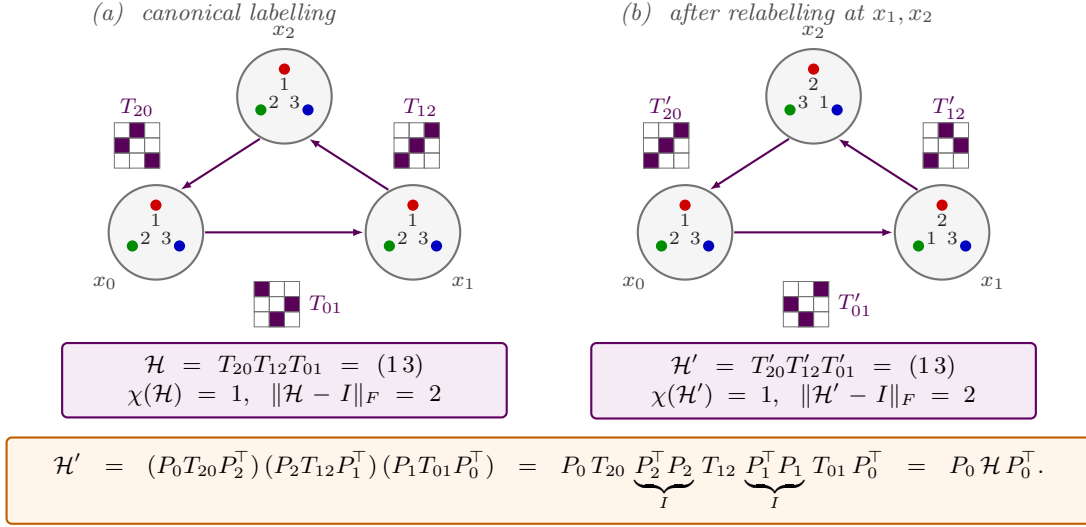


FIGURE 2. Each disc is a fibre $\hat{\mu}_{x_\alpha}$ of $N = 3$ ensemble members, depicted as red, green and blue dots, with small numerals giving the index labels assigned to those members at that vertex. Edge arrows carry transport matrices $T_{\alpha\beta}$, shown as 3×3 permutation glyphs (filled cell = unit entry). Panel (a) uses the canonical labelling at every vertex. Panel (b) relabels members at x_1 by $P_1 = (12)$ and at x_2 by $P_2 = (123)$, leaving the basepoint x_0 fixed. The colours stay put because the underlying ensemble does, while the glyph patterns differ between panels because each edge matrix is read against different conventions at its endpoints. The strip below shows that intermediate relabellings cancel pairwise in the loop product, leaving only conjugation by P_0 at the basepoint. With $P_0 = I$ here, $\mathcal{H}' = \mathcal{H}$ holds exactly. Conjugacy-class invariants (χ , eigenvalues, $\|\cdot - I\|_F$) are therefore unchanged by any relabelling, as recorded in the result boxes.

- (B) Assignment limit regime. Here we send $\varepsilon \downarrow 0$, under the standing hypothesis that the linear assignment problem on each edge of interest has a unique minimiser with a uniform suboptimality gap. Concretely, the jet-metric cost should single out a unique optimal permutation on each edge, with every competitor costlier by at least $\Delta_\star > 0$. Under this gap condition, Lemma 18 records the exponential concentration of $\Pi^{(\varepsilon)}$ around that permutation as $\varepsilon \downarrow 0$.
- (C) Exact assignment regime. Rather than taking a limit, one solves the linear assignment exactly on each edge, for instance by a Hungarian solver or the like, and obtains permutation matrices $T_{\alpha\beta} \in \mathbf{S}_N$ directly. Loop products then lie in $\mathbf{S}_N$ and admit the usual group interpretation. The global correspondence statements of Theorem 13 and Corollary 19 hold exactly in this regime.

Throughout, relabelling ensemble members at a vertex is modelled by conjugation by a permutation matrix, and we reserve group language for regime (C).

3.3. Discrete Connection Invariant Under Permutations. Adjacent inputs are connected by transporting their empirical fibres, a problem of statistical mechanics in the spirit of Jaynes. Among all couplings between the two fibres the one of maximum entropy is selected, subject to conservation of total probability and to a fixed average cost of matching jets under Definition 9. The resulting Gibbs distribution describes the most probable correspondence of ensemble members. When the temperature parameter tends to zero and the underlying linear assignment has a unique minimiser with a gap, the coupling concentrates near a deterministic matching (regime (B)). These edge maps compose along paths, forming a discrete connection whose loop product measures holonomy.

Definition 10 (Edge coupling and discrete connection). Let $(\mathcal{X}, G_h)$ be a connected neighbourhood graph over the feature space, with oriented edges $x_\alpha \rightarrow x_\beta$. For each edge we solve the entropic OT problem between the empirical fibres with the jet metric as ground cost and with uniform marginals. Denote its unique minimiser by $\Pi_{\alpha\beta}^{(\varepsilon)}$ and set

$$T_{\alpha\beta} := N \Pi_{\alpha\beta}^{(\varepsilon)},$$

a doubly stochastic operator. A discrete connection on the ensemble bundle is a choice of edge maps $\{T_{\alpha\beta}\}$ defined up to relabelling of ensemble members at vertices, that is, up to conjugation by permutation matrices P_α satisfying $T_{\alpha\beta} \sim P_\beta T_{\alpha\beta} P_\alpha^\top$.

Remark 11 (Gauge invariance under relabelling). Relabelling the models within the ensemble at a vertex x_α corresponds to permuting the basis of its fibre (see Figure 2). If this permutation is represented by a matrix P_α , then the edge map transforms as $T_{\alpha\beta} \mapsto P_\beta T_{\alpha\beta} P_\alpha^\top$. When we compose edge maps around a loop based at x_0 , the holonomy operator $\mathcal{H}_\gamma = T_{\alpha_{m-1}\alpha_m} \cdots T_{\alpha_0\alpha_1}$ transforms by conjugation, $\mathcal{H}_\gamma \mapsto P_0 \mathcal{H}_\gamma P_0^\top$. Consequently, any quantity that depends only on the conjugacy class of $\mathcal{H}_\gamma$ is insensitive to how ensemble members are labelled at intermediate vertices, and at the basepoint it depends on labels only through a single global permutation.

Remark 12 (Reversal convention). Definition 10 specifies edge maps only on oriented edges $x_\alpha \rightarrow x_\beta$, whereas for the reversed edge we adopt $T_{\beta\alpha} := T_{\alpha\beta}^\top$. In regime (C) this makes each T_e a permutation matrix, which is orthogonal, thus $T_{\beta\alpha} = T_{\alpha\beta}^{-1} = T_{\alpha\beta}^\top$. In regime (A) the transpose convention keeps the pair $(T_{\alpha\beta}, T_{\beta\alpha})$ consistent with the pushforward/pullback duality of the Gibbs coupling and is what our numerical pipeline implements (Appendix A). Throughout, I denotes the $N \times N$ identity matrix.

Theorem 13 (Holonomy and global correspondence). *Assume we are in the exact assignment regime (C). Let $G = (V, E)$ be a connected graph. For each oriented edge $(\alpha, \beta) \in E$ suppose we are given a transport matrix $T_{\alpha\beta} \in S_N$, and assume the reversal relation $T_{\beta\alpha} = T_{\alpha\beta}^{-1} = T_{\alpha\beta}^\top$.*

For a based loop $\gamma = (\alpha_0, \alpha_1, \dots, \alpha_L = \alpha_0)$ define its loop product by

$$\mathcal{H}_\gamma := T_{\alpha_{L-1}\alpha_L} \cdots T_{\alpha_0\alpha_1} \in S_N.$$

Then the following are equivalent:

- (1) *There exist permutations $P_\alpha \in S_N$ such that for every edge $(\alpha, \beta) \in E$ one has*

$$T_{\alpha\beta} = P_\beta P_\alpha^\top.$$

- (2) *$\mathcal{H}_\gamma = I$ for every based loop γ in G .*

If the equivalent conditions hold, then $\{P_\alpha\}$ is unique up to right multiplication by a single global permutation. Concretely, if $\{P_\alpha\}$ and $\{\tilde{P}_\alpha\}$ both satisfy $T_{\alpha\beta} = P_\beta P_\alpha^\top = \tilde{P}_\beta \tilde{P}_\alpha^\top$ for all (α, β) , then there exists $Q \in S_N$ such that $\tilde{P}_\alpha = P_\alpha Q$ for all $\alpha \in V$.

Proof. Assume (1). For any loop $\gamma = (\alpha_0, \dots, \alpha_L = \alpha_0)$, substituting $T_{\alpha_k\alpha_{k+1}} = P_{\alpha_{k+1}} P_{\alpha_k}^\top$ into the loop product (applied right to left) gives

$$\mathcal{H}_\gamma = \left(P_{\alpha_L} P_{\alpha_{L-1}}^\top \right) \cdots \left(P_{\alpha_1} P_{\alpha_0}^\top \right) = P_{\alpha_0} \left(\prod_{k=1}^{L-1} P_{\alpha_k}^\top P_{\alpha_k} \right) P_{\alpha_0}^\top = P_{\alpha_0} I P_{\alpha_0}^\top = I,$$

which is (2).

Assume (2). Fix a root vertex $\alpha_\star \in V$ and set $P_{\alpha_\star} := I$. For any vertex $\alpha \in V$ choose a path $\pi = (\alpha_\star = \beta_0, \beta_1, \dots, \beta_m = \alpha)$ and define the path product in the same right-to-left order as the loop product,

$$P_\alpha := T_{\beta_{m-1}\beta_m} \cdots T_{\beta_0\beta_1}.$$

If π' is another path from α_* to α , then π followed by the reverse of π' is a loop based at α_* . By (2) its loop product is I , and using $T_{\beta\alpha} = T_{\alpha\beta}^{-1}$ this implies the two path products agree, hence P_α is well-defined. For an edge $(\alpha, \beta) \in E$, take a path from α_* to α and append the edge (α, β) . By definition $P_\beta = T_{\alpha\beta} P_\alpha$, and therefore $T_{\alpha\beta} = P_\beta P_\alpha^\top$.

For uniqueness, if both $\{P_\alpha\}$ and $\{\tilde{P}_\alpha\}$ satisfy condition (1), then $\tilde{P}_\beta \tilde{P}_\alpha^\top = P_\beta P_\alpha^\top$ for every edge, and therefore $\tilde{P}_\alpha^\top P_\alpha = \tilde{P}_\beta^\top P_\beta$. Since G is connected, $\tilde{P}_\alpha^\top P_\alpha$ is the same permutation Q for every α , giving $\tilde{P}_\alpha = P_\alpha Q$ as claimed. $\square$

3.4. Holonomy Representation Of The Graph's Fundamental Group. The scalar diagnostic $h_{\text{op}}(\gamma) = \|\mathcal{H}_\gamma - I\|_F / N$ of Definition 22 is one projection of a richer object, the loop transport itself, which carries the structure of a representation of the graph's fundamental group. We make that structure explicit in what follows.

For two based loops γ_1, γ_2 at a fixed basepoint $x_0 \in V$, the concatenation $\gamma_1 \cdot \gamma_2$ traverses the edges of γ_1 first and those of γ_2 afterwards. Under the right-to-left convention of Definition 22, $\mathcal{H}_\gamma = T_{\alpha_{L-1}\alpha_L} \cdots T_{\alpha_0\alpha_1}$, the edges of γ_2 enter to the left of those of γ_1 , giving

$$\mathcal{H}_{\gamma_1 \cdot \gamma_2} = \mathcal{H}_{\gamma_2} \mathcal{H}_{\gamma_1}. \quad (12)$$

The inverse loop γ^{-1} satisfies $\mathcal{H}_{\gamma^{-1}} = \mathcal{H}_\gamma^{-1}$ using $T_{\beta\alpha} = T_{\alpha\beta}^\top$ in regime (C), and $\mathcal{H}_\gamma$ depends on γ only through its based homotopy class because backtracking a single edge contributes $T_{\alpha\beta}^\top T_{\alpha\beta} = I$ to the loop product.

Definition 14 (Holonomy representation). The map

$$\rho : \pi_1(G, x_0)^{\text{op}} \longrightarrow \text{End}(\mathbb{R}^N), \quad [\gamma] \longmapsto \mathcal{H}_\gamma,$$

is a monoid homomorphism in regime (A) and a group homomorphism into $\mathbf{S}_N$ in regime (C). A relabelling at the basepoint conjugates ρ by the corresponding permutation, letting ρ be well-defined up to conjugation.

For a connected graph $G = (V, E)$ with $|V|$ vertices and $|E|$ undirected edges, $\pi_1(G, x_0)$ is a free group on $r = |E| - |V| + 1$ generators, one for each edge outside a chosen spanning tree.

Definition 15 (Character spectrum). For a based loop γ and an integer $k \in \{1, \dots, N-1\}$, set

$$p_k(\gamma) := \frac{1}{N} \text{Tr}(\mathcal{H}_\gamma^k). \quad (13)$$

In regime (C) one has $p_k(\gamma) \in [-1, 1]$ and $p_k(\gamma) = \#\{\text{fixed points of } (\mathcal{H}_\gamma)^k\} / N$. The tuple $(p_1(\gamma), \dots, p_{N-1}(\gamma))$ is a class function on $\mathbf{S}_N$ and, by Newton's identities, determines the cycle type of $\mathcal{H}_\gamma$ and hence its conjugacy class. In regime (A) the $p_k(\gamma) \in \mathbb{C}$ are normalised power sums of the spectrum of the Markov operator $\mathcal{H}_\gamma$ with $|p_k(\gamma)| \leq 1$. In both regimes $p_k(\gamma)$ is invariant under basepoint gauge transformations.

Definition 16 (Abelianisation). The sign of the determinant,

$$\eta(\gamma) := \text{sign det } \mathcal{H}_\gamma \in \{\pm 1\},$$

defines a group homomorphism $\rho^{\text{ab}} : \pi_1(G, x_0) \rightarrow \mathbb{Z}_2$ which factors through the abelianisation $H_1(G, \mathbb{Z}_2) = (\mathbb{Z}_2)^r$. In regime (A) the same formula applies to the Markov operator $\mathcal{H}_\gamma$ and yields a group-to-monoid homomorphism with values in $\{-1, 0, +1\}$ on each based homotopy class.

The scalar diagnostic h_{op} is itself a projection of the first character. Expanding $\|\mathcal{H}_\gamma - I\|_F^2 = \text{Tr}(\mathcal{H}_\gamma^\top \mathcal{H}_\gamma) - 2 \text{Re Tr}(\mathcal{H}_\gamma) + N$ and using $\text{Tr}(\mathcal{H}_\gamma^\top \mathcal{H}_\gamma) = N$ in regime (C) and $\text{Tr}(\mathcal{H}_\gamma^\top \mathcal{H}_\gamma) \leq N$ in regime (A) yields

$$h_{\text{op}}(\gamma) = \frac{1}{\sqrt{N}} \sqrt{2(1 - \text{Re } p_1(\gamma))} \quad \text{in regime (C),} \quad (14)$$

with a bounded correction in regime (A). The summary of Definition 22 therefore records only the first power trace, while the higher p_k carry information invisible to h_{op} .

Theorem 17 (Conservation of character spectrum in regime (C)). *Assume we are in the exact assignment regime (C), and let $G = (V, E)$ be a connected graph with a spanning tree $\mathcal{T} \subset G$ and generating set $\{\gamma_1, \dots, \gamma_r\} \subset \pi_1(G, x_0)$ consisting of the $r = |E| - |V| + 1$ fundamental loops associated with the edges of $E \setminus \mathcal{T}$. The following are equivalent:*

- (1) $\mathcal{H}_\gamma = I$ for every based loop γ in G .
- (2) $p_k(\gamma_j) = 1$ for every $k \in \{1, \dots, N-1\}$ and every generator γ_j .
- (3) There exist vertex permutations $\{P_\alpha\}_{\alpha \in V} \subset S_N$ such that $T_{\alpha\beta} = P_\beta P_\alpha^\top$ for every edge $(\alpha, \beta) \in E$ and the family is unique up to right multiplication by a single global permutation.

Proof. The equivalence (1) $\iff$ (3) is Theorem 13. We show (1) $\iff$ (2).

(1) $\implies$ (2) is immediate: $\mathcal{H}_{\gamma_j} = I$ gives $\text{Tr}(\mathcal{H}_{\gamma_j}^k)/N = 1$ for every k .

(2) $\implies$ (1). Fix a generator γ_j . In regime (C), $\mathcal{H}_{\gamma_j} \in S_N$ has eigenvalues $\lambda_1, \dots, \lambda_N$ on the unit circle, implying $p_k(\gamma_j) = N^{-1} \sum_{i=1}^N \lambda_i^k$. By assumption $\sum_i \lambda_i^k = N$ for $k = 1, \dots, N-1$. Newton's identities [38] express the elementary symmetric polynomials $e_1, \dots, e_{N-1}$ in the λ_i as polynomials in $p_1, \dots, p_{N-1}$, and $\det \mathcal{H}_{\gamma_j} = \pm 1$ fixes e_N (every e_m is the same as for $\lambda_1 = \dots = \lambda_N = 1$). The characteristic polynomial of $\mathcal{H}_{\gamma_j}$ is therefore $(\lambda - 1)^N$, all eigenvalues equal 1, and a permutation matrix whose spectrum is $\{1\}$ must equal the identity. In other words, $\mathcal{H}_{\gamma_j} = I$ holds for every generator. By (12) and $\mathcal{H}_{\gamma^{-1}} = \mathcal{H}_\gamma^{-1}$, the set $\{\gamma \in \pi_1(G, x_0) : \mathcal{H}_\gamma = I\}$ is a subgroup of $\pi_1(G, x_0)$. Containing all γ_j , it coincides with $\pi_1(G, x_0)$, which gives (1). $\square$

Example. Theorem 17 reduces the conservativeness test of Theorem 13 from an a priori infinite family of loops to $r(N-1)$ power-traces on r fundamental loops. In Appendix A we consider a torus discretised on a 30×30 grid with $N = 10$, where this amounts to $901 \times 9 = 8109$ scalar equalities. For doubly stochastic $\mathcal{H}_\gamma$ at moderate ε , the determinant $\det \mathcal{H}_\gamma$ is not bounded away from zero. Long loops mix the Markov operator towards the rank-one stationary matrix $J = \frac{1}{N} \mathbf{1}\mathbf{1}^\top$, whose determinant vanishes, and the sign of $\det \mathcal{H}_\gamma$ is not a numerically resolvable ± 1 event in regime (A). We confirm this empirically in Appendix A, where the 30-edge non-contractible loops on the torus at $\varepsilon = \frac{1}{2}$ give $|\det \mathcal{H}_{\gamma_\alpha}| \sim 10^{-104}$ and $|\det \mathcal{H}_{\gamma_\beta}| \sim 10^{-112}$, while the power-trace tuple $(p_1, \dots, p_{N-1})$ remains informative across plaquettes. The topological content of η is therefore an invariant in regime (C), and its counterpart in regime (A) is recovered, within the assignment-limit sub-regime (B-gap), by the concentration of Lemma 18.

3.5. Regimes Of The Assignment Gap. For any $x, x' \in U$, where $U \subset \mathcal{X}$ is a fixed neighbourhood, we write $\{z_i(x)\}_{i=1}^N$ and $\{z_j(x')\}_{j=1}^N$ for the empirical jet supports with uniform weights, and set

$$C_{ij}(x, x') := d_Z^2((x, z_i(x)), (x', z_j(x'))).$$

Let $\sigma_{x, x'} \in S_N$ be any minimiser of the associated linear assignment problem, and define the edgewise gap

$$\Delta(x, x') := \min_{\tau \in S_N \setminus \{\sigma_{x, x'}\}} \sum_{i=1}^N C_{i, \tau(i)}(x, x') - \sum_{i=1}^N C_{i, \sigma_{x, x'}(i)}(x, x') \geq 0.$$

The behaviour of the entropic coupling $\Pi_{x, x'}^{(\varepsilon)}$ as $\varepsilon \downarrow 0$ is governed by the relationship between $\Delta(x, x')$ and ε . This relationship splits regime (B) of §3.2 into two sub-regimes.

Gapped assignment (B-gap). The linear assignment problem has a unique minimiser $\sigma_{x, x'}$ on each edge of interest, and $\Delta(x, x') \geq \Delta_\star > 0$ uniformly on $U \times U$ with $\Delta_\star$ independent of ε .

The standard exponential concentration [19] then applies, for some $K < \infty$ independent of $(x, x') \in U$,

$$\left\| \Pi_{x,x'}^{(\varepsilon)} - P_{\sigma_{x,x'}} \right\|_1 \leq K e^{-\Delta_*/\varepsilon},$$

where $P_{\sigma_{x,x'}}$ denotes the permutation matrix corresponding to $\sigma_{x,x'}$. Lemma 18 below establishes this rate, and Theorem 13 and Corollary 19 extend it to loop products.

Degenerate assignment (B-deg). Either the linear assignment admits multiple optima (the optimum lies on a face rather than a vertex of the Birkhoff polytope), or the edgewise gap vanishes along the temperature sweep, $\Delta(x, x') \rightarrow 0$ as $\varepsilon \downarrow 0$. In this case the uniform estimate above fails and the rates are polynomial in ε . The standard analysis of degenerate, entropic, linear programming [16, 19] yields

$$\bar{H} \left(\Pi_{x,x'}^{(\varepsilon)} \right) = O(\varepsilon), \quad \left\| N \Pi_{x,x'}^{(\varepsilon)} - P_{\sigma_{x,x'}} \right\|_F = O(\sqrt{\varepsilon}),$$

for the mean row entropy $\bar{H}$ and the Frobenius distance to any Hungarian optimum $P_{\sigma_{x,x'}}$. The limit point is then a distinguished interior element of the optimal face, typically not a permutation matrix.

3.6. Flat-Baseline Lemma And Holonomy Corollary. Within sub-regime (B-gap), entropic couplings concentrate exponentially around the optimal permutations, and loop products inherit the same rate. The lemma below records this concentration. All subsequent global correspondence statements, including Theorem 13 and Corollary 19, hold under that hypothesis. In sub-regime (B-deg) the limit transport is no longer a permutation, and the holonomy diagnostics of Definition 22 inherit the polynomial rates established above rather than the exponential one.

Lemma 18 (Calibration under a flat baseline). *Let $U \subset \mathcal{X}$ be a neighbourhood on which the empirical fibres coincide as unlabelled measures on jet space, $\hat{\mu}_x = \hat{\mu}_{x'}$ for all $x, x' \in U$, with common fibre size N and uniform marginals $a = b = \frac{1}{N} \mathbf{1}$. Assume that for each pair $(x, x') \in U \times U$ the linear assignment problem with ground cost induced by the jet metric has a unique minimising permutation $P_{x,x'}$, and that every competing permutation costs at least $\Delta_* > 0$ more than the optimum, uniformly in (x, x') . Then the entropy-regularised couplings $\Pi_{x,x'}^{(\varepsilon)}$ converge entrywise to $P_{x,x'}$ as $\varepsilon \downarrow 0$, and there exists $K < \infty$, independent of $(x, x') \in U$, such that*

$$\left\| \Pi_{x,x'}^{(\varepsilon)} - P_{x,x'} \right\|_1 \leq K e^{-\Delta_*/\varepsilon}.$$

Consequently $T_{x \rightarrow x'} := N \Pi_{x,x'}^{(\varepsilon)} \rightarrow P_{x,x'}$. For any fixed loop γ inside U , the loop product $\mathcal{H}_\gamma^{(\varepsilon)}$ converges to the loop product built from the limiting permutations $\{P_{x,x'}\}$. In particular, if that limiting loop product is the identity, then the holonomy summaries of Definition 22 vanish as $\varepsilon \downarrow 0$. The lemma thus serves as a calibration. Within a flat regime and at low temperatures, the Gibbs rule recovers a unique edgewise correspondence, while global coherence remains a distinct condition.

Proof. Form the cost matrix $C(x, x')$ from the jet metric. By hypothesis the unregularised assignment has a unique minimiser $P_{x,x'}$, and every competing permutation costs at least Δ_* more. The entropically penalised problem reads

$$\Pi_{x,x'}^{(\varepsilon)} \in \arg \min_{\Pi \in \mathcal{B}} \langle C(x, x'), \Pi \rangle + \varepsilon \text{KL} \left(\Pi \left\| \frac{1}{N^2} \mathbf{1} \mathbf{1}^\top \right\| \right), \quad (15)$$

where KL is the Kullback–Leibler divergence and $\mathcal{B}$ is the Birkhoff polytope. As $\varepsilon \downarrow 0$, the entropy penalty concentrates the minimiser on an extremal vertex of $\mathcal{B}$, and therefore on the unregularised optimum. Since the latter is unique, $\Pi_{x,x'}^{(\varepsilon)} \rightarrow P_{x,x'}$ entrywise [16, 19]. The entropic minimiser also admits the Gibbs form

$$\Pi_{ij}^{(\varepsilon)} = u_i \exp \left(- \frac{C_{ij}(x, x')}{\varepsilon} \right) v_j,$$

with positive scalings u, v enforcing the uniform marginals [20]. Decomposing along permutations shows that the total weight carried by any non-optimal permutation τ is suppressed by a factor at most $\exp(-\Delta_*/\varepsilon)$ relative to the optimal one. Uniform bounds on the scalings u, v over U and on the range of C then give

$$\|\Pi_{x,x'}^{(\varepsilon)} - P_{x,x'}\|_1 \leq K e^{-\Delta_*/\varepsilon}$$

for some $K < \infty$ independent of $x, x' \in U$ and for all sufficiently small $\varepsilon > 0$. This proves both the convergence and the exponential estimate. $\square$

The exponential rate of Lemma 18 is what characterises sub-regime (B-gap). The underlying structural requirement is uniform positivity of Δ_* across the ensemble. At every pair of neighbouring fibres, a distinguished pairing must exist whose jet-metric cost lies strictly below that of every competitor, by an amount that does not collapse as $\varepsilon \downarrow 0$.

Whether (B-gap) is realised must be checked empirically per experiment. Figure 3 contrasts the canonical sweep, where the entropic couplings concentrate only polynomially in ε and the rates are those of sub-regime (B-deg), with a tailored ensemble in which a fixed off-diagonal penalty enforces $\Delta_* > 0$ uniformly and the concentration becomes exponential as Lemma 18 predicts. Further calibrations appear in Appendix A. The general non-unique case, including the degenerate subcase, was treated in Carlier et al. [16].

Corollary 19 (Trivial holonomy yields a global labelling). *For exact assignment, assume regime (C), under which each edge transport is permutation-valued, $T_{\alpha\beta} \in S_N$, and satisfies $T_{\beta\alpha} = T_{\alpha\beta}^\top$. If $\mathcal{H}_\gamma = I$ for every loop γ in G , then there exist permutations $P_\alpha \in S_N$ such that for every edge $(\alpha, \beta) \in E$,*

$$T_{\alpha\beta} = P_\beta P_\alpha^\top.$$

In particular, for any two vertices $\alpha, \beta \in V$ and any path π from α to β , the induced path transport equals $P_\beta P_\alpha^\top$ and is independent of the chosen path.

For the assignment limit, assume sub-regime (B-gap) on a neighbourhood of interest, under which $T_{\alpha\beta} = N\Pi_{\alpha\beta}^{(\varepsilon)}$ converges to a permutation matrix as $\varepsilon \downarrow 0$ with uniform uniqueness gap $\Delta_ > 0$. If the limiting permutation transports satisfy $\mathcal{H}_\gamma = I$ on every loop of the graph, then the entropic loop products converge to the identity on each fixed loop and the scalar summaries in Definition 22 converge to zero at the exponential rate of Lemma 18. In sub-regime (B-deg) the same scalar summaries still converge whenever the limit transport is trivial, although the rate is polynomial in ε and the limit plan is in general not a permutation.*

Proof. The exact assignment statement follows immediately from Theorem 13. The assignment limit statement follows from Lemma 18 edgewise together with continuity of loop products under perturbations, as in Theorem 32. $\square$

To pass to a mesh approximation, we work on a shape-regular mesh $\mathcal{T}_h$ with maximum edge length $O(h)$ inside a geodesic ball of radius strictly below the injectivity radius of $(\mathcal{X}, g_\mathcal{X})$. Along each edge (x_α, x_β) form the entropic plan $\Pi_{\alpha\beta}^{(\varepsilon)}$ and the stochastic map $T_{\alpha \rightarrow \beta}$ as in Definition 10. Send the regularisation in the assignment regime jointly with the mesh and ensemble size, $\varepsilon = \varepsilon(h, N) \downarrow 0$, $h \downarrow 0$, and $N = N(h) \uparrow \infty$, in such a way that $\|\Pi_{\alpha\beta}^{(\varepsilon)} - P_{\alpha\beta}\|_1 \leq K e^{-\Delta_*/\varepsilon} = o(h)$ uniformly on edges, where $P_{\alpha\beta}$ is the optimal permutation supplied by Lemma 18. The tangent transport entering $d\mathbf{z}$ is approximated by the embedded Procrustes scheme, whose edgewise error is $O(h)$. The resulting ground costs and assignments are therefore C^1 in the endpoints to first order in h .

Remark 20 (General loops). Any contractible mesh loop of diameter $O(h)$ decomposes into $O(1)$ such triangles, and the same type of estimate carries over up to a constant when one compares loop products along neighbouring loops. For loops of fixed combinatorial length, refine the mesh and approximate the loop by a geodesic polygon. By continuity of edge transports with respect to the

vertices (Theorem 32), the resulting loop products converge to the loop product associated with the limiting edge transports along that polygon. Convergence to the identity requires an additional triviality condition on that limiting loop product, as in the exact assignment regime of Theorem 13 and Corollary 19.

At the level of indices, $\Pi_{x \rightarrow x'}$ is close to a permutation and pairs corresponding members at x and x' . At the level of features, its barycentric projection transports jets in $\mathcal{Z}$ coherently with the ground cost.

Proposition 21 (Permutation invariance). *The ensemble at each input consists of N models with no fixed ordering. A change of ordering amounts to applying a permutation matrix at that input. Under such a relabelling, the transport $\Pi_{x \rightarrow x'}$ transforms by conjugation with the relevant permutations. Any scalar built from these transport maps reflects the collective behaviour of the ensemble alone and is therefore invariant under relabelling. In particular, the connection itself is well defined up to such permutations, and every holonomy scalar derived from it is an intrinsic property of the unlabelled ensemble.*

Proof. Let P_σ, P_τ be the permutation matrices for the relabellings at x and x' respectively. Relabelling the ensemble at a vertex amounts to permuting its fibre basis. The OT cost is invariant under permutation of source and target atoms, and the optimal plan is carried to $\Pi'_{x \rightarrow x'} = P_\sigma \Pi_{x \rightarrow x'} P_\tau^\top$. The transfer operator $T_{x \rightarrow x'} = N \Pi_{x \rightarrow x'}$ acts on column distributions at x and produces column distributions at x' . Under the relabellings P_σ at the source and P_τ at the target it transforms as

$$T'_{x \rightarrow x'} = P_\tau T_{x \rightarrow x'} P_\sigma^\top,$$

in agreement with Definition 10.

Along a path $x_0 \rightarrow x_1 \rightarrow \dots \rightarrow x_m$, the transported product becomes

$$\begin{aligned} T'_{x_{m-1} \rightarrow x_m} \cdots T'_{x_0 \rightarrow x_1} = \\ P_{\sigma_m} T_{x_{m-1} \rightarrow x_m} P_{\sigma_{m-1}}^\top \cdots P_{\sigma_1} T_{x_0 \rightarrow x_1} P_{\sigma_0}^\top. \end{aligned}$$

Adjacent permutations $P_{\sigma_i}^\top P_{\sigma_i}$ cancel, leaving the overall product conjugate to the unpermuted holonomy. Since every scalar we consider is invariant under conjugacy, its value is unchanged by relabelling. The connection is therefore defined only up to permutation, while any holonomy scalar constructed from it is independent of the chosen ensemble labels. $\square$

3.7. Holonomy As A Loop Transport. Local edge transports match ensemble members between nearby inputs. Whether these matchings stay coherent across the graph is a separate question, addressed by following a path $\gamma = (x_0, \dots, x_L)$ and composing the edge maps in order, carrying the fibre at x_0 along the path and back to itself. When the transport is path-independent the resulting map is the identity, and any deviation records the curvature of the connection along γ .

Definition 22 (Holonomy operator and scalar summaries). For a path $\gamma = (x_0, x_1, \dots, x_L)$ define the loop transport

$$\mathcal{H}_\gamma := T_{x_{L-1} \rightarrow x_L} \cdots T_{x_0 \rightarrow x_1}.$$

If γ is a loop with $x_L = x_0$, then $\mathcal{H}_\gamma$ acts on distributions over the N atoms of $\hat{\mu}_{x_0}$. In regime (A) this is a Markov operator (doubly stochastic in our normalisation), and in regime (C) it is a permutation matrix.

Write the fibre atoms at x_0 as $\{z_i(x_0)\}_{i=1}^N \subset \mathcal{Z}$. Two scalar summaries serve in what follows,

$$\begin{aligned} \text{h}_{\text{op}}(\gamma) &:= \frac{1}{N} \|\mathcal{H}_\gamma - I\|_{\text{F}}, \\ \text{h}_{\text{bar}}(\gamma) &:= W_2(\hat{\mu}_{x_0}, \mathcal{H}_\gamma \# \hat{\mu}_{x_0}), \end{aligned} \tag{16}$$

where W_2 is taken with d_Z at x_0 . Here $\mathcal{H}_{\gamma\#}\hat{\mu}_{x_0}$ is interpreted by barycentric projection on the base fibre, namely the empirical measure

$$\mathcal{H}_{\gamma\#}\hat{\mu}_{x_0} := \frac{1}{N} \sum_{i=1}^N \delta_{\sum_{j=1}^N (\mathcal{H}_{\gamma})_{ij} z_j(x_0)}.$$

Both summaries are invariant under relabelling of ensemble members, since relabelling conjugates $\mathcal{H}_{\gamma}$ by a permutation matrix and hence permutes the family of barycentric images.

The invariant attached to a based loop is the operator $\mathcal{H}_{\gamma}$ itself, or equivalently its conjugacy class at the basepoint. The scalars $h_{\text{op}}(\gamma)$ and $h_{\text{bar}}(\gamma)$ are reductions of this object to a single real number. A finer gauge-invariant summary is the character spectrum of Definition 15. In regime (C) the tuple $(p_1(\gamma), \dots, p_{N-1}(\gamma))$ determines the cycle type of $\mathcal{H}_{\gamma} \in S_N$, and therefore its conjugacy class, in full. In regime (A) the same tuple encodes the normalised power sums of the spectrum of the Markov operator $\mathcal{H}_{\gamma}$ and, by the Newton correspondence, its characteristic polynomial. The identity (14) shows that h_{op} sees only the first power trace. By Theorem 17, evaluating $p_1, \dots, p_{N-1}$ on any generating set of $\pi_1(G, x_0)$ is a finite sufficient test for conservativeness in regime (C).

These quantities are small when the edgewise correspondences compose nearly consistently around the loop, and grow as inconsistency accumulates. They are not comparing the loop length to a chord length. They compare the composite loop transport to the identity action on the base fibre. If the loop product equals the identity operator, then both summaries vanish. In regime (A), any deviation from zero is most naturally read as a diagnostic of inconsistency of local matchings under the chosen entropic transport rule, since the connection is not group-valued.

Proposition 23 (Flat baselines). *Assume that for every edge $(\alpha, \beta) \in E$ the cost matrix used to compute $\Pi_{\alpha\beta}^{(\varepsilon)}$ is the same matrix C_0 , independent of (α, β) , and that the entropic OT problem with uniform marginals for C_0 has the unique solution $\Pi_0^{(\varepsilon)}$. Then $\Pi_{\alpha\beta}^{(\varepsilon)} = \Pi_0^{(\varepsilon)}$ for every edge, and therefore*

$$T_{\alpha\beta} = N\Pi_{\alpha\beta}^{(\varepsilon)} = T_0 \quad \text{for all } (\alpha, \beta) \in E,$$

where $T_0 := N\Pi_0^{(\varepsilon)}$ is doubly stochastic. Consequently, for any loop γ of length L , one has

$$\mathcal{H}_{\gamma} = T_0^L.$$

In particular, if $T_0 = I$, as in an exact assignment baseline where the unique minimiser is the identity permutation on every edge, then $\mathcal{H}_{\gamma} = I$ for every loop γ , and all holonomy summaries vanish.

Proof. Edge independence of the cost and uniqueness of the entropic solution imply $\Pi_{\alpha\beta}^{(\varepsilon)} = \Pi_0^{(\varepsilon)}$ for each edge. The identity $\mathcal{H}_{\gamma} = T_0^L$ follows by multiplying the constant edge transport along the loop. The final claim follows by setting $T_0 = I$. $\square$

Proposition 24 (Stability under refinement). *Fix $\varepsilon > 0$. Consider a sequence of paths $\gamma^{(m)} = (x_0^{(m)}, \dots, x_L^{(m)})$ of fixed length L , with $x_i^{(m)} \rightarrow x_i$ for each i as $m \rightarrow \infty$. Assume the ground costs induced by d_Z depend smoothly on the vertices in a neighbourhood of the limiting path $\gamma = (x_0, \dots, x_L)$, and that the fibre atoms remain separated throughout, free of collisions. Then the corresponding loop transports $\mathcal{H}_{\gamma^{(m)}}$ converge to $\mathcal{H}_{\gamma}$, and the scalar summaries $h_{\text{op}}(\gamma^{(m)})$ and $h_{\text{bar}}(\gamma^{(m)})$ converge to $h_{\text{op}}(\gamma)$ and $h_{\text{bar}}(\gamma)$.*

In particular, if a refinement scheme produces polygonal paths whose vertices converge in this sense to a limiting polygonal path, then the associated loop transports and summaries converge along the refinement.

Proof. By strict convexity of the entropic objective, $\Pi_{x \rightarrow x'}^{(\varepsilon)}$ is the unique minimiser of (15) and depends smoothly on the entries of the cost matrix. Smooth dependence of d_Z on the vertices implies $C(x_{i-1}^{(m)}, x_i^{(m)}) \rightarrow C(x_{i-1}, x_i)$ entrywise, hence $T_{x_{i-1}^{(m)} \rightarrow x_i^{(m)}} \rightarrow T_{x_{i-1} \rightarrow x_i}$ for each i .

Write $A_i^{(m)} := T_{x_{i-1}^{(m)} \rightarrow x_i^{(m)}}$ and $A_i := T_{x_{i-1} \rightarrow x_i}$. The telescoping estimate in the proof of Theorem 32 gives

$$\left\| A_L^{(m)} \cdots A_1^{(m)} - A_L \cdots A_1 \right\| \leq \sum_{k=1}^L \left\| A_k^{(m)} - A_k \right\|,$$

for any induced ℓ^1 (or ℓ^∞) operator norm, since each $A_i^{(m)}$ and A_i is doubly stochastic and therefore has operator norm 1. Thus $\mathcal{H}_{\gamma^{(m)}} \rightarrow \mathcal{H}_\gamma$. Continuity of the Frobenius norm and of W_2 with respect to the barycentric pushforward in Definition 22 yields convergence of h_{op} and h_{bar} . $\square$

4. A DISCRETE STOKES LAW FOR S_N -VALUED CONNECTIONS

The face holonomies of §3 assemble into a global identity, expressing the boundary holonomy of a simply connected region as a tree-ordered product of its face holonomies each conjugated to a common basepoint, in the lineage of the recursive plaquette aggregation of lattice gauge theory [32, 34, 41, 52] and providing the discrete counterpart of Stokes's theorem for the connection $\{T_{\alpha\beta}\}$. The law restates a standard fact of lattice gauge theory for transports valued in the symmetric group, and the work here lies in that specialisation and in the abelian invariants drawn from it rather than in the identity itself. Applying $\eta = \text{sign det}$ produces the closed-surface sum rule that underlies the topological signature on the torus, of the same form as the $\mathbb{Z}_2$ cocycle observable of Wegner [50], and a doubly stochastic version of the same identity carries a reversibility deficit through a product of squared face determinants.

Throughout this section $D \subset G$ is a simply connected 2-cell subcomplex of the mesh $\mathcal{T}_h$ with face set $F(D)$, basepoint x_0 at a vertex of ∂D , and the oriented boundary ∂D read counter-clockwise. For each face $c \in F(D)$ we write $\mathcal{H}_c$ for the face holonomy of Definition 22 based at a chosen corner v_c of c , and fix a spanning tree $\mathcal{T}_D$ of the vertices of D rooted at x_0 . For each face c , let π_c be the unique $\mathcal{T}_D$ -path from x_0 to v_c , and set $g_c := \mathcal{H}_{\pi_c}$, with values in S_N in regime (C) and in the doubly stochastic Birkhoff polytope B_N in regime (A). A tree-compatible face ordering of $F(D)$ is any enumeration $(c_1, \dots, c_n)$ such that, at each step k , the path π_{c_k} lies inside the boundary of the still-unpeeled region $D_k := D \setminus \bigcup_{j < k} c_j$. Such an ordering exists for every pair $(\mathcal{T}_D, D)$, and for a rectangular D with the right-then-up tree it may be taken as columns left-to-right, top-to-bottom within each column.

Theorem 25 (Non-abelian discrete Stokes law, regime (C)). *In the exact-assignment regime (C), for any simply connected $D \subset G$, any spanning tree $\mathcal{T}_D$, and any tree-compatible ordering $(c_1, \dots, c_n)$ of $F(D)$,*

$$\mathcal{H}_{\partial D} = \prod_{k=1}^n g_{c_k}^{-1} \mathcal{H}_{c_k} g_{c_k} \quad (17)$$

in S_N , where the product is read left-to-right and the opposite-group composition of Definition 14 places the factor labelled by c_1 at the far left. In particular, the right-hand side does not depend on the choice of $(\mathcal{T}_D, v_c, \text{ordering})$ beyond the constraint of tree-compatibility.

Proof. Induction on $n = |F(D)|$.

Base case $n = 1$. A single face c_1 with $v_{c_1} = x_0$ gives $g_{c_1} = I$ and $\mathcal{H}_{\partial D} = \mathcal{H}_{c_1}$. For $v_{c_1} \neq x_0$ one has $\partial D = \pi_{c_1} \cdot \partial c_1 \cdot \pi_{c_1}^{-1}$ in $\pi_1(G, x_0)^{\text{op}}$, hence by (12) $\mathcal{H}_{\partial D} = g_{c_1}^{-1} \mathcal{H}_{c_1} g_{c_1}$.

Inductive step. Suppose the identity holds for every simply connected subcomplex with fewer than n faces, and let D have $n \geq 2$ faces with a tree-compatible ordering $(c_1, \dots, c_n)$. The peeled region $D_n := D \setminus c_n$ is itself simply connected by the tree-compatibility of the ordering, and

$(c_1, \dots, c_{n-1})$ is a tree-compatible ordering of D_n . Let σ be the part of ∂c_n that is shared with ∂D_n and σ^c the remaining portion lying on ∂D , each read with the orientation induced by c_n . Choose v_{c_n} to be the vertex at which σ^c starts along ∂D . A direct computation in $\pi_1(G, x_0)^{\text{op}}$ yields

$$\partial D = \partial D_n \cdot (\pi_{c_n} \cdot \partial c_n \cdot \pi_{c_n}^{-1}),$$

after cancelling $\sigma \cdot \sigma^{-1}$ along the shared edges. Tree-compatibility ensures that π_{c_n} lies in ∂D_n , and no further homotopy across faces of D_n is needed. Applying ρ and (12) gives

$$\mathcal{H}_{\partial D} = (g_{c_n}^{-1} \mathcal{H}_{c_n} g_{c_n}) \mathcal{H}_{\partial D_n},$$

with the new factor multiplied on the left, the cumulative ordering matching $(c_1, \dots, c_{n-1}, c_n)$ read left-to-right in the matrix product. The inductive hypothesis applied to $\mathcal{H}_{\partial D_n}$ completes the proof. Gauge independence of the right-hand side follows from Definition 14. Under a relabelling P_{x_0} at the basepoint, both sides transform by $P_{x_0}(\cdot)P_{x_0}^T$, and under a different tree-compatible choice the intermediate changes amount to insertions of $(\pi\pi^{-1})$ factors that cancel in the product. $\square$

Taking the sign of the determinant on both sides of (17) yields the abelianised form, for which the gauge factors drop out and no tree ordering is needed.

Corollary 26 (Abelian discrete Stokes law, regime (C)). *For any simply connected $D \subset G$ in regime (C),*

$$\eta(\partial D) = \prod_{c \in F(D)} \eta(c), \quad (18)$$

in $\{\pm 1\}$.

Proof. The determinant is multiplicative and conjugation-invariant, and therefore $\det(g_c^{-1} \mathcal{H}_c g_c) = \det \mathcal{H}_c$. Applying $\det$ to (17) gives $\det \mathcal{H}_{\partial D} = \prod_c \det \mathcal{H}_c$. Each factor lies in $\{\pm 1\}$ because $\mathcal{H}_c \in \mathbf{S}_N$, and taking signs yields the claim. $\square$

Corollary 27 (Closed-surface sum rule). *Let Σ be a closed oriented 2-surface triangulated by G with face set $F(\Sigma)$. Then, for any connection in regime (C),*

$$\prod_{c \in F(\Sigma)} \eta(c) = +1. \quad (19)$$

Proof. Writing $\eta(c) = \prod_{e \in \partial c} \text{sign det } T_e$ with e oriented by c and using the identity $\det T_{\beta\alpha} = \det T_{\alpha\beta}^{-1} = (\det T_{\alpha\beta})^{-1}$ in regime (C),

$$\prod_{c \in F(\Sigma)} \eta(c) = \prod_{e \in E(\Sigma)} (\text{sign det } T_e)^{k_e},$$

where k_e counts the two oriented traversals of e by its two adjacent faces. On a closed oriented surface every edge is traversed exactly once in each direction, and the two contributions are $\text{sign det } T_e$ and $\text{sign det } T_e^{-1} = \text{sign det } T_e$. Their product equals $(\text{sign det } T_e)^2 = +1$, and the total product is $+1$. $\square$

On the torus $\Sigma = T^2$, the closed-surface sum rule does not detect the two independent non-contractible loops $\gamma_\alpha, \gamma_\beta$ of $\pi_1(T^2) \cong \mathbb{Z}^2$. Those loops bound no region, and their holonomies carry a genuinely topological signature.

Corollary 28 (Topological signature on the torus). *In regime (C), let $\gamma_\alpha, \gamma_\beta$ be the two non-contractible loop generators of $\pi_1(T^2, x_0) \cong \mathbb{Z}^2$. The pair*

$$(\eta_\alpha, \eta_\beta) := (\eta(\gamma_\alpha), \eta(\gamma_\beta)) \in \mathbb{Z}_2 \times \mathbb{Z}_2$$

is invariant under all gauge relabellings at the basepoint and under continuous deformations of $\gamma_\alpha, \gamma_\beta$ within their based homotopy classes. It is invisible to every diagnostic evaluated on contractible loops and to every local summary of the field. Any loop γ homologous to $a\gamma_\alpha + b\gamma_\beta \pmod{2}$ satisfies $\eta(\gamma) = \eta_\alpha^a \eta_\beta^b$, and the four classes $(\eta_\alpha, \eta_\beta) \in \{\pm 1\}^2$ therefore partition the space of regime-(C) connections on T^2 into gauge-invariant sectors that no contractible observable can separate, in bijection with the elements of $H^1(T^2, \mathbb{Z}_2)$.

Proof. Gauge invariance is the content of Definition 16 restricted to $\pi_1(T^2, x_0)$. For contractible blindness, any contractible loop γ bounds a simply connected region $D \subset T^2$, and by (18) $\eta(\gamma) = \prod_{c \in F(D)} \eta(c)$, a product of face invariants insensitive to the representation on non-contractible generators. For the homological transfer, η factors through $\rho^{\text{ab}} : \pi_1(T^2) \rightarrow \mathbb{Z}_2$ (Definition 16), and therefore $\eta(a\gamma_\alpha + b\gamma_\beta) = \eta_\alpha^a \eta_\beta^b \pmod{2}$. $\square$

The codomain $\mathbb{Z}_2 \times \mathbb{Z}_2$ is the first $\mathbb{Z}_2$ -cohomology of the torus, and $\Sigma(\rho) := \eta \circ \rho^{\text{ab}}$ is a class in it.

Proposition 29 (Topological signature as a class in $H^1(T^2, \mathbb{Z}_2)$). *Let G be a 2-cell discretisation of the torus T^2 , $x_0 \in V$ a basepoint, and $\rho : \pi_1(G, x_0)^{\text{op}} \rightarrow \mathbb{S}_N$ a regime-(C) connection. The assignment*

$$\Sigma(\rho) := \eta \circ \rho^{\text{ab}} \in \text{Hom}(\pi_1(T^2, x_0), \mathbb{Z}_2) \cong H^1(T^2, \mathbb{Z}_2) \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \quad (20)$$

defines a class in the first $\mathbb{Z}_2$ -cohomology of T^2 , invariant under basepoint relabelling and under continuous deformation of generators within their based homotopy classes. Twisting any base connection by an odd permutation along the α - or β -seam toggles the corresponding component of Σ , and Σ is therefore surjective onto the four classes of $H^1(T^2, \mathbb{Z}_2)$.

Proof. Invariance under basepoint relabelling and homotopic deformation is the content of Corollary 28. The universal coefficient theorem applied to the free abelian group $H_1(T^2, \mathbb{Z}) \cong \mathbb{Z}^2$ gives $H^1(T^2, \mathbb{Z}_2) \cong \text{Hom}(\mathbb{Z}^2, \mathbb{Z}_2) \cong \mathbb{Z}_2 \times \mathbb{Z}_2$. Surjectivity is the seam-twist construction. Multiplying every α -seam edge transport on the left by a fixed odd permutation $\sigma \in \mathbb{S}_N$ multiplies $\rho(\gamma_\alpha)$ on the left by σ and leaves $\rho(\gamma_\beta)$ unchanged, hence flips η_α and preserves η_β . The same on the β -seam flips η_β and preserves η_α , and composing the two reaches every $(a, b) \in \mathbb{Z}_2 \times \mathbb{Z}_2$ from any base connection. $\square$

Remark 30 (Relation to spin structures). The group $H^1(T^2, \mathbb{Z}_2)$ acts simply transitively on the set of spin structures of the torus [8], so the four sectors of Σ stand in bijection with those spin structures once a base point of the torsor is fixed. The correspondence is one of sets carrying an action of the same group and carries no further structure. The construction supplies no quadratic refinement of the intersection form, no Arf invariant, and no Dirac operator, and Σ therefore records the cohomology class alone and not a spin structure in the geometric sense.

The four classes are exhibited concretely in Appendix A, with the marker ensemble producing the trivial class $(+1, +1)$ and the canonical connection together with two of its $\mathbb{S}_N$ -twists realising the remaining classes $(-1, -1)$, $(+1, -1)$, $(-1, +1)$.

In regime (A) the non-abelian Stokes identity does not lift verbatim to doubly stochastic transports, because the reversal relation weakens from $T_{\beta\alpha} = T_{\alpha\beta}^{-1}$ to $T_{\beta\alpha} = T_{\alpha\beta}^\top$ and interior edges no longer cancel at the operator level. They cancel up to a transpose-squared determinant factor, and the corresponding closed-surface identity carries a reversibility deficit.

Proposition 31 (Regime-(A) Stokes deficit). *In regime (A) with doubly stochastic edge transports T_e and the reversal $T_{\beta\alpha} = T_{\alpha\beta}^\top$, for any simply connected $D \subset G$,*

$$\prod_{c \in F(D)} \det \mathcal{H}_c = \det \mathcal{H}_{\partial D} \cdot \prod_{e \in E^\circ(D)} (\det T_e)^2, \quad (21)$$

where $E^\circ(D)$ is the set of interior edges of D . Consequently, on a closed oriented surface Σ ,

$$\prod_{c \in F(\Sigma)} \det \mathcal{H}_c = \prod_{e \in E(\Sigma)} (\det T_e)^2 \in [0, 1], \quad (22)$$

with the right-hand side equal to 1 if and only if every T_e is a permutation matrix. The quantity $\prod_{e \in E(\Sigma)} (\det T_e)^2$ is a gauge-invariant, global reversibility defect of the doubly stochastic connection.

Proof. Using $\det T_{\beta\alpha} = \det T_{\alpha\beta}^\top = \det T_{\alpha\beta}$ and the multiplicativity of $\det$ along each face boundary,

$$\prod_{c \in F(D)} \det \mathcal{H}_c = \prod_{e \in E(D)} (\det T_e)^{m_e},$$

where m_e is the total number of face traversals of e regardless of orientation. For interior edges $m_e = 2$, each traversed by both adjacent faces, and for boundary edges $m_e = 1$. Collecting terms yields (21). On a closed surface every edge is interior, whence (22). Each $\det T_e$ lies in $[-1, 1]$ by Hadamard's inequality for doubly stochastic matrices, with $|\det T_e| = 1$ iff T_e is a permutation matrix [12]. Squaring gives the stated range. $\square$

In the assignment limit $\varepsilon \downarrow 0$ within sub-regime (B-gap), $|\det T_e| \rightarrow 1$ exponentially in $1/\varepsilon$ (Lemma 18), and (22) recovers the regime-(C) sum rule of Corollary 27. In sub-regime (B-deg) the deficit remains bounded away from 1 by a polynomial in ε , and its size is itself a diagnostic of how strongly the doubly stochastic limit fails to be permutation-valued.

Each concept above has a concrete finite counterpart. A neighbourhood graph on sampled inputs furnishes empirical fibres and their measures, the jet metric provides edgewise alignment, and a transport problem is solved on each edge. The discrete approximation viewpoint and stability under mesh refinement are taken up in §5, with the construction collected in compact form at the start of Appendix A for the experiments that follow.

5. DISCRETE APPROXIMATIONS

The continuum constructions of §2 and §3 assume access to the manifold, its tangent geometry, and the true fibres. In practice, we observe only a finite ensemble at finitely many inputs. We therefore replace the manifold by a graph, the smooth transport by entropic couplings with a Gibbs kernel, and the continuum holonomy by a loop operator on faces. Standard conditions then ensure that these discrete objects approximate their continuum analogues.

5.1. Parallel Transport Over Graphs Via Gibbs Couplings. Let $\mathcal{T}_h$ be a shape-regular mesh of $\mathcal{X}$ with maximal edge length $h > 0$ and write $V_h = \{x_\alpha\}$ for its vertices and E_h for its undirected edges. At each vertex x_α form the empirical measure $\hat{\mu}_{x_\alpha}$ on the jet space $\mathcal{Z}$ from the ensemble $\{f_{\theta_i}\}_{i=1}^N$ as in §2 [see 13, 44]. For an edge $(x_\alpha, x_\beta) \in E_h$, define the ground cost as

$$c_{\alpha\beta}((x_\alpha, z), (x_\beta, z')) := d_{\mathcal{Z}}^2((x_\alpha, z), (x_\beta, z')),$$

where $d_{\mathcal{Z}}$ is the jet metric of Definition 9, using ambient alignment or Levi-Civita transport along the edge. Let $\Pi_{\alpha\beta}^{(\varepsilon)}$ be the optimal coupling between $\hat{\mu}_{x_\alpha}$ and $\hat{\mu}_{x_\beta}$ with uniform marginals $1/N$ and parameter $\varepsilon > 0$. As before, the associated transfer operator, $T_{\alpha\beta} := N\Pi_{\alpha\beta}^{(\varepsilon)}$, is doubly stochastic and admits the Gibbs representation $\Pi_{\alpha\beta, ij}^{(\varepsilon)} = u_i \exp(-C_{ij}/\varepsilon) v_j$ for the cost matrix C and positive scalings u, v [20]. In the assignment regime $\varepsilon \downarrow 0$, and under the uniqueness and uniform gap hypothesis from Lemma 18, $T_{\alpha\beta}$ converges to the optimal permutation matrix [11, 16, 19]. The discrete connection along the oriented edge $x_\alpha \rightarrow x_\beta$ acts on probability row vectors $p \in \Delta^{N-1}$ by right multiplication, $p \mapsto pT_{\alpha\beta}$, and on column vectors by left multiplication with $T_{\alpha\beta}^\top$.

When we appeal to the flat-baseline theory, we take $\varepsilon = \varepsilon(h, N) \downarrow 0$ as $h \rightarrow 0$ and $N \rightarrow \infty$, and we further assume that the underlying assignments fall in sub-regime (B-gap) of §3.5, with $\Pi_{\alpha\beta}^{(\varepsilon)}$

within numerical tolerance of the unique permutation from Lemma 18. When the sweep instead lies in sub-regime (B-deg), the plan converges polynomially to a non-permutation limit, and projecting to the nearest permutation is an additional modelling step rather than a consequence of small ε . In experiments we either set $\varepsilon = 0$ and solve the linear assignment directly, or operate at a moderate ε and report whichever regime applies. The calibrated comparisons of Appendix A make this dichotomy explicit.

5.2. Discrete Holonomy On Faces And Sections. Local transports align ensemble members between adjacent inputs, and the global question is whether these local alignments compose consistently around a closed path, and whether a coherent representative can be selected at each vertex over the entire graph. Holonomy on a face tests this by asking whether transport around a small cycle returns to its starting configuration, just as parallel-transporting a vector around a triangle on a curved surface may rotate it. A canonical representative field along a spanning tree provides the second test, with its dependence on the chosen tree itself a witness of curvature.

Let c be a two-cell of $\mathcal{T}_h$ with ordered boundary $(x_{\alpha_0}, x_{\alpha_1}, \dots, x_{\alpha_m} = x_{\alpha_0})$. Its loop transport (holonomy) operator is

$$\mathcal{H}_h(c) := T_{\alpha_{m-1}\alpha_m} \cdots T_{\alpha_0\alpha_1},$$

which acts on probability distributions over the N ensemble indices at the base vertex x_{α_0} . The matrix $\mathcal{H}_h(c)$ equals the identity if the chosen loop product closes exactly at that base. In the permutation-valued regime this coincides with path independence in the usual sense. A convenient scalar summary follows from (16) as

$$h_h(c) := \frac{1}{N} \|\mathcal{H}_h(c) - I\|_F,$$

which vanishes precisely when the loop closes. To compare faces of different sizes and across mesh refinements, we also consider the area-rescaled quantity

$$\kappa_h(c) := \frac{h_h(c)}{\text{area}(c)}, \quad (23)$$

used below as a curvature surrogate. Both h_h and κ_h are invariant under any relabelling of the ensemble at intermediate vertices, i.e. a change of indexing of the N independently trained models within a fibre, represented by a permutation P_x on the canonical basis at x . Such a change sends each edge map to $T_{uv} \mapsto P_v T_{uv} P_u^\top$ and hence $\mathcal{H}_h(c)$ to $P_{x_{\alpha_0}} \mathcal{H}_h(c) P_{x_{\alpha_0}}^\top$. The Frobenius distance to the identity is therefore a class function and unchanged by relabelling.

To construct representatives across the graph, fix a base vertex x_0 and label its fibre by the one-hot distributions $\{e_j\}_{j=1}^N \subset \Delta^{N-1}$. Choose a spanning tree $\mathcal{T} \subset \mathcal{T}_h$. For each vertex x_α , transport these labels along the unique tree path to obtain

$$p^{(j)}(x_\alpha) := e_j \prod_{(u \rightarrow v) \in \text{path}(x_0 \rightarrow x_\alpha)} T_{uv},$$

with the product ordered along the path. Project each $p^{(j)}(x_\alpha)$ to jet space by the barycentric map over the atoms of $\hat{\mu}_{x_\alpha}$,

$$z^{(j)}(x_\alpha) := \sum_{i=1}^N p_i^{(j)}(x_\alpha) z_i(x_\alpha).$$

The meshwise mean section is then

$$s_h^*(x_\alpha) := \frac{1}{N} \sum_{j=1}^N z^{(j)}(x_\alpha). \quad (24)$$

Because every T_{uv} has uniform marginals, the average $\frac{1}{N} \sum_j p^{(j)}(x_\alpha)$ is the uniform distribution for all x_α . Consequently $s_h^*(x_\alpha)$ is simply the barycentre of the fibre at x_α , independent of the chosen

tree and of any relabelling at the base, and it serves as a gauge-free reference field. Curvature is not seen in this mean but in the spread of the individual sections $z^{(j)}$ and, more canonically, in the loop operator $\mathcal{H}_h$.

We use the barycentric projection to induce loop actions in jet space. A one-hot atom at the base is transported around a loop by $\mathcal{H}_h(c)$, mapped to jet space by the rule above, and compared to its starting jet using the base metric. Since edge maps are only defined up to conjugation by permutations, any displacement produced by such a loop is unaffected by relabellings at intermediate vertices and changes at the base only by a global permutation of indices. When holonomy is trivial on the faces that close the tree, every $z^{(j)}$ propagated along $\mathcal{T}$ is independent of the chosen tree up to a fixed global permutation. When holonomy is non-trivial, the residual dependence on the tree is the discrete witness of non-integrability. The barycentric projection is standard in OT literature for turning couplings into feature maps (see, for example, [44]).

5.3. Existence, Stability, And Convergence. For admissible inputs the edge couplings exist and are unique, they vary gently under small perturbations of the jets, and under refinement the loop quantities settle to the expected continuum behaviour. Each of these follows from the formulation already in place, and no special numerics are required.

For a fixed edge (x_α, x_β) , the coupling $\Pi_{\alpha\beta}^{(\varepsilon)}$ is the maximum entropy distribution on the product fibre subject to the uniform marginal constraints and a prescribed average cost, namely the minimiser of (15). Introducing Lagrange multipliers $\{a_i\}$ and $\{b_j\}$ for the two marginal constraints gives the Gibbs form

$$\Pi_{\alpha\beta,ij}^{(\varepsilon)} = \frac{1}{Z} \exp\left(\frac{a_i + b_j - C_{\alpha\beta,ij}}{\varepsilon}\right) = u_i K(C_{\alpha\beta})_{ij} v_j, \quad K(C)_{ij} := e^{-C_{ij}/\varepsilon},$$

with positive scalings $u = \exp(a/\varepsilon)$ and $v = \exp(b/\varepsilon)$ fixed by the uniform marginals. Strict convexity of the free energy under entropy penalisation on the transportation polytope and feasibility of the constraints yield existence and uniqueness. The map $(C_{\alpha\beta}, a, b) \mapsto \Pi_{\alpha\beta}^{(\varepsilon)}$ is smooth. The Jacobian of the marginal constraints with respect to (a, b) is invertible on the positive orthant, and the Implicit Function Theorem therefore applies, with $\Pi_{\alpha\beta}^{(\varepsilon)}$ depending smoothly on $C_{\alpha\beta}$ [44]. In practice one may recover (u, v) by iterative scaling, but this algorithmic detail plays no role in the analysis.

Theorem 32 (Existence and stability of the discrete connection). *For each edge (x_α, x_β) , the problem (15) with uniform marginals has a unique minimiser $\Pi_{\alpha\beta}^{(\varepsilon)}$, and $T_{\alpha\beta} := N\Pi_{\alpha\beta}^{(\varepsilon)}$ is doubly stochastic. The maps $T_{\alpha\beta}$ and every loop product $\mathcal{H}_\gamma$ depend continuously on the entries of the ground cost matrices. Under the fibrewise gauge action by permutations, $T_{\alpha\beta}$ transforms by conjugation at its endpoints and $\mathcal{H}_\gamma$ transforms by conjugation at the chosen base. All scalar summaries used below are class functions and therefore gauge invariant. No expansion near the identity is asserted without a fixed gauge.*

Proof. Existence and uniqueness are as above. Smooth dependence of $\Pi_{\alpha\beta}^{(\varepsilon)}$ on $C_{\alpha\beta}$ follows from the Gibbs dual and the Implicit Function Theorem [44], and smooth dependence of $C_{\alpha\beta}$ on the jets then gives continuous dependence of $T_{\alpha\beta}$ on the data. To pass from edges to loops, write a loop as an ordered product and compare two families $\{A_i\}$ and $\{\tilde{A}_i\}$. For any submultiplicative operator norm,

$$\left\| \prod_{i=1}^m A_i - \prod_{i=1}^m \tilde{A}_i \right\| \leq \sum_{k=1}^m \left\| A_1 \cdots A_{k-1} \right\| \left\| A_k - \tilde{A}_k \right\| \left\| \tilde{A}_{k+1} \cdots \tilde{A}_m \right\|.$$

Taking the induced ℓ^1 -norm (or ℓ^∞ -norm), each doubly stochastic factor has operator norm 1, hence

$$\left\| \prod_{i=1}^m A_i - \prod_{i=1}^m \tilde{A}_i \right\| \leq \sum_{k=1}^m \|A_k - \tilde{A}_k\|.$$

This telescoping estimate transfers continuity from edges to loop products and therefore to the scalar summaries $h_h(c)$ and $\kappa_h(c)$. Gauge covariance under fibrewise permutations is the standard conjugation law already recorded. Because $h_h(c) = \frac{1}{N} \|\mathcal{H}_h(c) - I\|_F$ depends only on the conjugacy class, it is gauge invariant, as is $\kappa_h(c)$. The final statement on expansions follows from the absence of a fixed gauge. $\square$

On flat baselines with a unique assignment and a uniform gap hypothesis, Lemma 18 applies. If $\varepsilon(h) \downarrow 0$ along a refinement, then $\Pi_{\alpha\beta}^{(\varepsilon(h))} \rightarrow \frac{1}{N} P_{\alpha\beta}$ entrywise at an exponential rate in $1/\varepsilon(h)$ [19]. Hence $T_{\alpha\beta} \rightarrow P_{\alpha\beta}$, and for any fixed loop (in particular, for any fixed face cycle c of bounded combinatorial length) the loop product built from $T_{\alpha\beta}$ converges to the loop product built from the limiting permutations $P_{\alpha\beta}$. In particular, if the limiting permutation loop products are trivial on the collection of loops under consideration, then $h_h(c) \rightarrow 0$ and $\kappa_h(c)$ remains bounded. In the general smooth case at fixed $\varepsilon > 0$ we do not claim a Taylor expansion near the identity. What we use is continuity and uniform bounds, i.e. small perturbations of the jets produce small perturbations of the costs, hence small changes in the edge maps, and the telescoping bound then gives uniform estimates on $\mathcal{H}_h(c)$ on faces of bounded area. This ensures stability of the diagnostics (23) under data and mesh perturbations and yields tightness of area-weighted sums. For the mean section s_h^* in (24), the barycentric projection is 1-Lipschitz in the ℓ^1 weights with respect to the jet metric, the transported one-hots remain probability vectors, and the family is equibounded. Finally, after piecewise interpolation an Arzelà–Ascoli argument gives the stated convergence.

Remark 33 (On small loops). An expansion near the identity for loop products requires a smooth action on an ambient space together with a fixed gauge. In the discrete, label-valued permutation limit no Taylor expansion is available. Any curvature estimate should therefore be formulated in jet space via the barycentric map under explicit regularity and separation assumptions, a direction we develop elsewhere. The scalars h_h and κ_h are employed as diagnostics over a finite mesh, with $h_h(c)$ measuring loopwise deviation from flatness and $\kappa_h(c)$ its area-rescaled counterpart, the latter behaving stably under refinement.

6. CLOSING THOUGHTS

We have developed a discrete S_N -gauge theory on 2-cell complexes whose fibres are empirical measures on N states and whose connection is the entropic OT coupling between neighbouring fibres. The conjugacy invariants of the loop transport, namely the character spectrum and the abelianisation $\eta = \text{sign det}$, descend to gauge-invariant scalars on $\pi_1(G, x_0)$. The non-abelian discrete Stokes law of Theorem 25 expresses the boundary holonomy of a simply connected region as a tree-ordered product of conjugated face holonomies, and its abelianisation produces the closed-surface sum rule of Corollary 27. The latter detects only contractible information on a closed oriented surface. The non-contractible homology of Σ is read instead by the character spectrum on the generators of $\pi_1(G, x_0)$, and on T^2 collapses to the $\mathbb{Z}_2 \times \mathbb{Z}_2$ topological signature of Corollary 28, realised by Proposition 29 as a class in the first $\mathbb{Z}_2$ -cohomology $H^1(T^2, \mathbb{Z}_2)$ of the torus.

A conservative connection, that is one for which ρ is trivial on every cycle, admits a global labelling of fibre coordinates over $\mathcal{X}$, by Theorem 13. When ρ fails to be trivial, the obstruction is a gauge-invariant property of the connection, recorded by the conjugacy invariants of $\mathcal{H}_\gamma$ over a finite set of generators of $\pi_1(G, x_0)$ via Newton’s identities (Theorem 17). Among the conjugacy invariants the abelianisation η is the coarsest and the most stable, valued in $\{\pm 1\}$ on every loop and accessible to a single sign computation per face. The character spectrum sharpens the picture,

with the higher power traces $\text{Re } p_k$ for $k \geq 2$ carrying information that no scalar contraction of p_1 retains. The Frobenius gap $h_{\text{op}} = \sqrt{2(1 - \text{Re } p_1)/N}$ is one such contraction, and its empirical insensitivity to $p_2, \dots, p_{N-1}$ on the canonical ensemble (Appendix A) is consistent with the algebra.

The abelian identities of §4 are confirmed on explicit connections. The closed-surface sum rule $\prod_c \eta(c) = +1$ and the abelian Stokes identity $\eta(\partial D) = \prod_{c \in F(D)} \eta(c)$ hold under exact assignment, and the four sectors of $\mathbb{Z}_2 \times \mathbb{Z}_2$ are reached on one base geometry by twisting the connection along the non-contractible seams, with a marker connection at the trivial class. These checks are reported in Appendix A, and the genus- g form of the signature is worked out in Appendix B.

The construction depends on the data of an empirical measure on a finite set at each vertex and a cost between neighbouring fibres, and it admits instantiations far beyond the example presented here. Configurations of an N -state lattice model, S_N -coloured spin foams, finite-state lattice models in the lineage of Baez [9], Kogut [34], Wegner [50], Wilson [52], and finite-orientation crystal grain fields are direct instances, one of which is worked through in Appendix B. The Stokes deficit of Proposition 31 extends the framework to doubly stochastic edge transports, and the $\varepsilon \downarrow 0$ behaviour of the deficit is governed by the gap dichotomy of Lemma 18.

The continuum limit $(h, \varepsilon) \rightarrow 0$ along $\varepsilon/h^2 \rightarrow \lambda$, in which the discrete connection should reproduce a smooth connection on $\mathcal{X}$ with curvature 2-form involving the antisymmetrised covariant derivative of the jet covariance, is a natural target for a separate paper. The extension of Proposition 29 to closed orientable surfaces of higher genus, where $H^1(\Sigma_g, \mathbb{Z}_2) \cong \mathbb{Z}_2^{2g}$, is carried out on a cell model of Σ_g in Appendix B. The behaviour of the character spectrum under refinement of G , and the consistency of the abelian Stokes law under mesh refinement, are both constrained by the discrete approximation results of §5, although a quantitative rate is not yet established.

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APPENDIX A. SYNTHETIC EXPERIMENTS ON THE TORUS

The construction takes as data an ensemble of N trained models, a finite set of basepoints carrying a neighbourhood graph $G_h = (V_h, E_h)$, the jet-metric weights $\lambda_{\text{val}}, \lambda_{\text{jac}}$ of Definition 9, a regularisation temperature $\varepsilon \geq 0$, and a collection of loops γ on which to read diagnostics, for instance face boundaries in a mesh or short cycles in a k -nearest neighbour graph. From these it returns the discrete connection $\{T_{\alpha\beta}\}$, the loop products $\mathcal{H}_\gamma$, and the conjugacy-invariant scalars $h_{\text{op}}(\gamma)$ and $h_{\text{bar}}(\gamma)$ of Definition 22, and in regime (C) it recovers a global labelling $\{P_\alpha\}$ by Theorem 13 when every loop product is the identity.

The procedure runs in four stages. At each basepoint x_α one forms the empirical fibre and its measure as in Definition 8, with tangent frames and the edgewise alignment estimated locally when $\mathcal{X}$ is a point cloud, and the jet metric gives the ground cost $C_{\alpha\beta,ij} = d_{\mathcal{Z}}^2((x_\alpha, z_i(x_\alpha)), (x_\beta, z_j(x_\beta)))$ on each oriented edge. The edge transport is the rescaled entropic coupling $T_{\alpha\beta} = N \Pi_{\alpha\beta}^{(\varepsilon)}$ in regime (A), doubly stochastic and acting as a Markov operator on the N ensemble indices, or the exact assignment permutation $T_{\alpha\beta} \in \mathbf{S}_N$ in regime (C). Composing edge maps along a loop $\gamma = (x_0, \dots, x_L = x_0)$ gives the loop product $\mathcal{H}_\gamma = T_{x_{L-1} \rightarrow x_L} \cdots T_{x_0 \rightarrow x_1}$ acting on distributions over the base fibre. The two scalars of Definition 22 follow, the operator deviation h_{op} reading the distance of $\mathcal{H}_\gamma$ from the identity in Frobenius norm and the barycentric displacement h_{bar} reading how far the loop action moves the fibre in jet space, and over many small loops they give a field on the base.

We instantiate the procedure on the standard torus, with an ensemble of small trained sections as fibres serving only to produce an empirical measure on N states at each vertex. We record below the base geometry, the ensemble, the connection, the diagnostics, the baselines that fix the numerical floor, and the verifications of the abelian identities, and the accompanying code reproduces every figure and number [24].

The base space is the standard embedded torus $\Phi : \mathbb{T}^2 \rightarrow \mathbb{R}^3$ with major radius $R = 1$ and minor radius $r = 1$,

$$\Phi(u, v) = ((R + r \cos v) \cos u, (R + r \cos v) \sin u, r \sin v), \quad (u, v) \in [0, 2\pi)^2.$$

An orthonormal tangent frame $E(u, v) \in \mathbb{R}^{2 \times 3}$ is formed by normalising the coordinate partials and stacking them as rows, and it projects ambient vectors to intrinsic components throughout. The induced area weight is $r(R + r \cos v) du dv$. The target field is $g(u, v) = \sin(2u + 4 \sin v) + \frac{1}{2} \cos(8v - u)$, and the ambient target $h^*(x) = g(\Phi^{-1}(x))$ is well defined because Φ is an embedding.

The dataset draws $M = 1000$ points with u_k, v_k independent and uniform on $[0, 2\pi)$, mapped to $X_k = \Phi(u_k, v_k)$ with labels $y_k = g(u_k, v_k)$. A multilayer perceptron $f_\theta : \mathbb{R}^3 \rightarrow \mathbb{R}$ with one hidden layer of width 10 and ReLU activations, $f_\theta(x) = W_2 \sigma(W_1 x + b_1) + b_2$, is trained to approximate h^* by minimising the mean squared error with the Adam optimiser at step size 10^{-2} , batch size 64, and 200 epochs, without explicit regularisation. Repeating for fixed seeds $s = 0, \dots, 9$ produces an ensemble of $N = 10$ sections. At any evaluation point the predictions $\{f_{\theta(s)}(x)\}$ and ambient gradients $\{\nabla_x f_{\theta(s)}(x)\}$ form a fibre, with the gradients projected onto the intrinsic frame to give jets.

Evaluation is on a regular 30×30 grid of angles, mapped to points $x_\alpha = \Phi(u_\alpha, v_\alpha)$ with the frame $E(u_\alpha, v_\alpha)$ precomputed. For each model and grid point we record the scalar prediction and the ambient gradient, project the latter to two intrinsic components by Gram–Schmidt on the columns of $D\Phi$, and store the jet $z = (y, g_1, g_2)$. The empirical fibre at x_α is the multiset $\{z_i(x_\alpha)\}_{i=1}^N$. As a local reference unrelated to transport we also record the ensemble prediction standard deviation σ_y and the Frobenius norm of the 2×2 gradient covariance at each point.

Tangent frames rotate across the surface, so jets are compared in a common gauge. For neighbours (x_α, x_β) we form $E_\alpha E_\beta^\top$ and take its closest $SO(2)$ rotation $R_{\alpha\beta}$ by singular value decomposition, carrying $g(x_\beta)$ into the frame at x_α before any distance is measured. Value and gradient are

placed on comparable scales by global dispersions s_y and s_g computed over all models and points and held fixed. The ground cost between jets on an edge is

$$C_{ab} = \left(\frac{y_\alpha^{(a)} - y_\beta^{(b)}}{s_y} \right)^2 + \left\| \frac{g_\alpha^{(a)} - R_{\alpha\beta} g_\beta^{(b)}}{s_g} \right\|_2^2.$$

The entropic coupling with uniform marginals $a = b = \frac{1}{N} \mathbf{1}$ is the Gibbs solution of (15) at temperature ε in the kernel $K_{ij} = e^{-C_{ij}/\varepsilon}$, computed in the log domain through the dual potentials

$$\varphi \leftarrow \varphi + \log \frac{1}{N} - \log \sum_j e^{(\varphi + \psi - C_{ij})/\varepsilon}, \quad \psi \leftarrow \psi + \log \frac{1}{N} - \log \sum_i e^{(\varphi + \psi - C_{i\cdot})/\varepsilon},$$

iterated until the marginal residual falls below 10^{-9} or 1000 iterations are reached. The doubly stochastic edge map is $T_{\alpha \rightarrow \beta} = N P^*$. Temperature is set adaptively as $\varepsilon = \alpha \text{median}(C)$ with $\alpha \in [2 \times 10^{-2}, 10^{-1}]$, which keeps short-edge transport near the identity, and a small offset guards the degenerate case of exact ties.

Parallel transport is read on a plaquette $\gamma = (x_0, x_1, x_2, x_3, x_0)$ by composing the four edge maps,

$$\mathcal{H}_\gamma = T_{3 \rightarrow 0}^\top T_{2 \rightarrow 3}^\top T_{1 \rightarrow 2}^\top T_{0 \rightarrow 1}^\top,$$

where the transpose turns the stored coupling, indexed with source first, into a left-acting push-forward on column vectors. Two scalar summaries follow. The Frobenius gap is $h_{\text{op}}(\gamma) = \frac{1}{N} \|\mathcal{H}_\gamma - I\|_F$ of Definition 22. The barycentric displacement pushes the base jets through $\mathcal{H}_\gamma$ to $\tilde{z}_i = \sum_j (\mathcal{H}_\gamma)_{ij} z_j(x_0)$ and reports $\frac{1}{N} \sum_i \|D^{-1}(\tilde{z}_i - z_i)\|_2$ with $D = \text{diag}(s_y, s_g, s_g)$, a surrogate for the W_2 quantity in (16). An area-rescaled gap $\kappa(\gamma) = h_{\text{op}}(\gamma)/\text{area}(\gamma)$ with $\text{area} = r(R + r \cos v) \Delta u \Delta v$ stabilises comparison across meshes and loop sizes. Both summaries are invariant under relabelling at intermediate vertices and under rotation of tangent frames, and change only by a global permutation at the base. As $\varepsilon \downarrow 0$ the edge maps approach permutations and integrable alignments give $\mathcal{H}_\gamma \approx I$, while as $\varepsilon \uparrow \infty$ they tend to $J = \frac{1}{N} \mathbf{1}\mathbf{1}^\top$, whence $h_{\text{op}} \rightarrow \sqrt{N-1}/N$.

Computing these summaries on every unit plaquette gives a loop inconsistency field over the surface. To check that this field is not a restatement of local spread, we scatter it against σ_y and the gradient covariance and report Spearman correlations with bootstrap intervals from 500 resamples, finding that the field tracks neither local baseline closely (Table 1). Stability is monitored by sweeping ε , by varying the loop side length, and by refining the mesh, where fields at face centres are compared across resolutions by periodic bilinear interpolation in (u, v) under area-weighted norms, with convergence declared when sup-norm and L^2 differences fall below 10^{-3} . Gauge invariance is confirmed by replacing E with QE for smooth $Q \in SO(2)$, reparameterisation invariance by the warp $(u, v) \mapsto (u + \eta \sin v, v)$ with pullback, and locality by reconstructing from sliding windows, all reproducing the field within 10^{-10} .

Two flat baselines fix the numerical floor against which canonical values are read. Setting every edge map to the identity forces $\mathcal{H} = I$ on every face and both summaries vanish, which validates the loop composition independently of the optimiser. Driving each edge to the identity matching by a strong off-diagonal penalty at $\varepsilon = 10^{-4}$ leaves a field mean holonomy of 5.73×10^{-13} . The opposite extreme $K \equiv \mathbf{1}\mathbf{1}^\top$ gives $T = J$, hence $\mathcal{H} = J$ and $h_{\text{op}} = \sqrt{N-1}/N$. The identity baseline fixes a global indexing and so breaks permutation invariance, and it is a calibration rather than evidence of coherence. Any non-trivial loop summary at moderate ε with no tie break therefore reflects non-integrability of the connection and not a coding artefact or local dispersion.

Vanishing holonomy is also reachable without an identity prior in the solver. Each model receives a tiny marker constant over the surface, a scalar δ_s added to its output and a two-vector v_s added to its intrinsic gradient, with δ_s centred and equally spaced and the v_s equally spaced on the circle. The markers cancel when a model is paired with itself and add a fixed positive margin

$$\left(\frac{\delta_a - \delta_b}{s_y} \right)^2 + \left\| \frac{v_a - v_b}{s_g} \right\|_2^2 \geq m_0 > 0$$

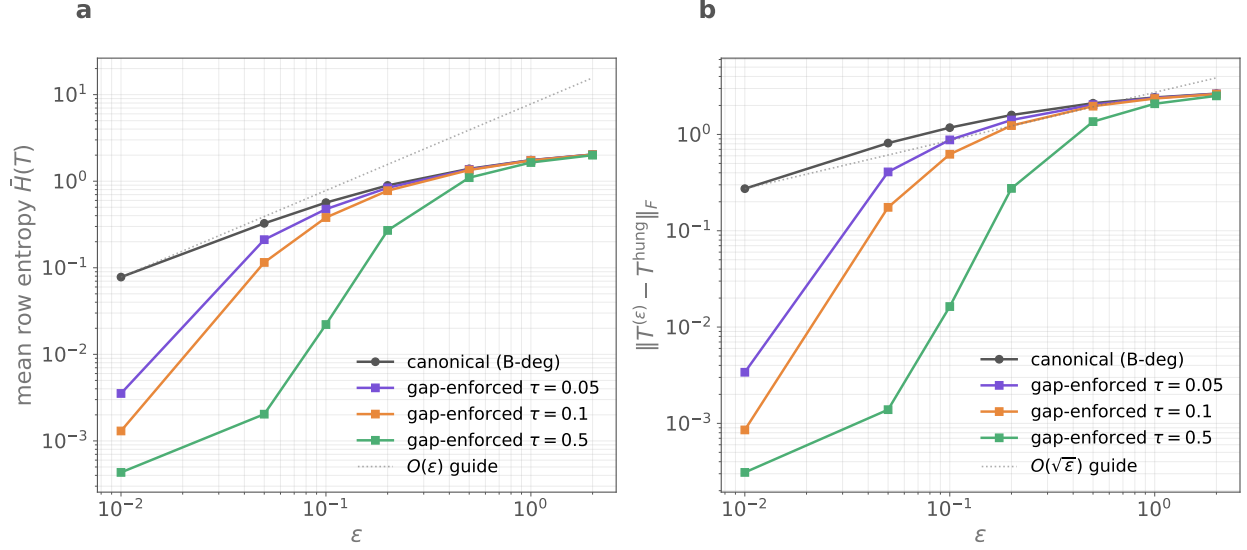


FIGURE 3. Concentration of the entropic edge maps as $\varepsilon \downarrow 0$, on log-log axes. **(a)** mean row entropy $\bar{H}(T)$ and **(b)** Frobenius distance to the exact permutation $\|T^{(\varepsilon)} - T^{\text{hung}}\|_F$. The canonical sweep tracks the polynomial guides $O(\varepsilon)$ and $O(\sqrt{\varepsilon})$ of the degenerate sub-regime (B-deg), while the ensembles enforced by the three gaps with $\tau \in \{0.05, 0.1, 0.5\}$ fall faster than any power, the exponential rate of sub-regime (B-gap) in Lemma 18.

when distinct models are paired. For large enough marker scales the margin makes self-pairing the unique optimum on every edge, so the edge maps and loop products are the identity to roundoff. The loop at the centre then reads $h_{\text{op}} = 2.55 \times 10^{-11}$, the field mean is 1.24×10^{-10} , and the gauge and reparameterisation checks pass within the same tolerances. On the same curved base, when the learned correspondence admits a global labelling, holonomy vanishes, and the non-zero values of the canonical run therefore reflect the connection rather than the surface.

The entropic and the exact regimes can be compared on identical edge costs. Exact linear assignment by the Hungarian algorithm returns the permutation $\sigma^* = \arg \min_{\sigma} \sum_i C_{i\sigma(i)}$ and the permutation edge map T^{hung} , with the same loop composition as the entropic case. At $\varepsilon = \frac{1}{2}$ the entropic field is spatially smooth with mean 0.274 and range $[0.220, 0.300]$, while the Hungarian field is wider and noisier with mean 0.307 and range $[0.000, 0.447]$, since hard assignment amplifies discrete structure and a small change in the cost can flip the optimal permutation. As $\varepsilon \downarrow 0$ the entropic maps approach a permutation, measured by the mean row entropy $\bar{H}(T) = -\frac{1}{N} \sum_i \sum_j p_j \log p_j$ of the normalised rows and by the Frobenius distance to T^{hung} . Both fall monotonically, with $\bar{H} = O(\varepsilon)$ and the distance $O(\sqrt{\varepsilon})$, the polynomial rates of the degenerate sub-regime (B-deg) of §3.5. The edgewise gap $\hat{\Delta}$ on this sweep is not bounded away from zero and scales with ε , so the uniform-gap hypothesis of Lemma 18 does not hold and the exponential rate is recovered only on a gap-enforced ensemble, both shown in Figure 3. Permutation gauge invariance is confirmed directly by drawing an independent random relabelling at each grid point and recomputing the field, where five draws agree with the baseline to 3×10^{-17} in L^2 and to $\rho = 1.0000$ in rank, as Proposition 21 requires.

The scalar h_{op} reads only the first power trace through (14), and the character spectrum $(p_1, \dots, p_{N-1})$ of Definition 15 carries the rest. Across the 29×29 interior plaquettes the correlation of $\text{Re } p_k$ with $\text{Re } p_1$ falls from 1.00 at $k = 1$ to 0.58 at $k = 9$, so the higher characters hold gauge-invariant information that no contraction of p_1 retains. On the two non-contractible

TABLE 1. Numerical verification on the periodic 30×30 torus discretisation with $N = 10$. Entropic couplings at $\varepsilon = \frac{1}{2}$ except where exact assignment in regime (C) is stated. Full data and code at [24].

Check	Result
Closed-surface sum rule $\prod_c \eta(c)$, regime (C)	+1 (490 at +1, 410 at -1)
Abelian Stokes $\eta(\partial D) = \prod_c \eta(c)$ on 500 rectangles, regime (C)	500/500 agree
Signature for canonical, α -twist, β -twist, marker	$(-1, -1), (+1, -1), (-1, +1), (+1, +1)$
Character correlation $\text{Re } p_k$ with $\text{Re } p_1$	1.00 ($k=1$) to 0.58 ($k=9$)
Non-contractible $ \det \mathcal{H}_{\gamma_\alpha} $ and $ \det \mathcal{H}_{\gamma_\beta} $	$\sim 10^{-104}$ and $\sim 10^{-112}$
Loop field against prediction std and gradient covariance	Spearman $ \rho \lesssim 0.15$ and ≈ 0.5
Permutation gauge invariance over 5 relabellings	$L^2 \approx 3 \times 10^{-17}$, $\rho = 1.0000$
Identity baseline and marker ensemble field mean h_{op}	5.7×10^{-13} and 1.2×10^{-10}
Entropic and Hungarian field mean h_{op}	0.274 and 0.307

generators at $\varepsilon = \frac{1}{2}$ both loops give $p_k \approx 1/N$, the 30-edge Markov product having mixed to J , and the determinants $|\det \mathcal{H}_{\gamma_\alpha}| \sim 10^{-104}$ and $|\det \mathcal{H}_{\gamma_\beta}| \sim 10^{-112}$ place the abelianisation below any numerical threshold in regime (A). The abelianisation is a regime-(C) invariant, and the identities it enters are therefore measured under exact assignment.

The four statements on which Sections 3 to 4 rest are verified in regime (C), where exact assignment makes the abelian identities hold exactly, and are collected in Table 1. On the periodic 30×30 discretisation all 900 plaquette signs multiply to $\prod_c \eta(c) = +1$, with 490 positive and 410 negative, confirming (19) on a closed surface while each $\eta(c)$ ranges over both signs. The abelian Stokes identity $\eta(\partial D) = \prod_{c \in F(D)} \eta(c)$ holds with 500/500 sign agreement on 500 random rectangles up to 29×29 plaquettes. The canonical ensemble carries the signature $(\eta_\alpha, \eta_\beta) = (-1, -1)$. Multiplying the stored edge transport on each ν -seam crossing by the odd permutation $\sigma = (01)$ flips one component at a time and yields $(+1, -1)$ and $(-1, +1)$, and the marker ensemble above sits at $(+1, +1)$, so all four sectors of $\mathbb{Z}_2 \times \mathbb{Z}_2$ are realised on one base geometry and the signature is blind to every contractible observable.

APPENDIX B. PERMUTATION CONNECTIONS ON A HIGHER-GENUS SURFACE

The construction asks only for an empirical measure on a finite set at each vertex and a cost between neighbouring fibres, and the trained ensemble of Appendix A is one source of that data among many. A polycrystal furnishes another. Grain orientations valued in a finite point group $G \subset SO(3)$, sampled at the sites of a planar section and labelled by N representative orientations, give an N -state fibre at each site, with the misorientation between neighbouring grains as the cost. A grain boundary that winds around a non-contractible cycle of a periodic sample carries permutation holonomy of the kind exhibited below, and its abelianisation is the same $\mathbb{Z}_2$ class. The holonomy, the character spectrum, and the abelianisation are defined from the fibres and the cost alone.

We exhibit the construction directly on a closed orientable surface of genus g , where the abelian theory of §4 extends without change and the signature occupies the full group $H^1(\Sigma_g, \mathbb{Z}_2) \cong \mathbb{Z}_2^{2g}$. Take the standard cell model of Σ_g with one vertex x_0 , the $2g$ oriented edges $a_1, b_1, \dots, a_g, b_g$, and a single face attached along the relation

$$\partial c = \prod_{i=1}^g [a_i, b_i] = a_1 b_1 a_1^{-1} b_1^{-1} \cdots a_g b_g a_g^{-1} b_g^{-1}.$$

A regime-(C) connection assigns a permutation to each generator, $\rho(a_i) = A_i$ and $\rho(b_i) = B_i$ in S_N , and the face holonomy is the image of the relation,

$$\mathcal{H}_c = \prod_{i=1}^g [A_i, B_i].$$

Each commutator satisfies $\text{sign det}[A_i, B_i] = (\text{sign det } A_i)^2 (\text{sign det } B_i)^2 = +1$, so $\eta(c) = +1$ on the single face and the closed-surface sum rule of Corollary 27 holds identically, for every choice of connection. The signature is the tuple of generator parities

$$(\eta(a_1), \eta(b_1), \dots, \eta(a_g), \eta(b_g)) = (\text{sign det } A_1, \dots, \text{sign det } B_g) \in \mathbb{Z}_2^{2g},$$

and choosing each generator even or odd reaches every one of the $2^{2g} = 4^g$ classes of $H^1(\Sigma_g, \mathbb{Z}_2)$.

For $g = 2$ and $N = 5$, take

$$A_1 = (0\ 1), \quad B_1 = (0\ 1\ 2), \quad A_2 = (1\ 2), \quad B_2 = (2\ 3),$$

with parities $(\eta(a_1), \eta(b_1), \eta(a_2), \eta(b_2)) = (-1, +1, -1, -1)$. The face holonomy evaluates to the 3-cycle $\mathcal{H}_c = (0\ 1\ 3)$ up to the composition convention of Definition 14, of cycle type $(3, 1, 1)$ and $\text{sign det } \mathcal{H}_c = +1$ as the sum rule requires. Multiplying the b_1 -edge transport on the left by the odd permutation $\sigma = (0\ 1)$ flips $\eta(b_1)$ from $+1$ to -1 and leaves the other three components fixed, the genus- g form of the seam twist of Proposition 29. Ranging the four generator parities independently realises all 16 classes of $H^1(\Sigma_2, \mathbb{Z}_2)$ on the one cell model, with $\text{sign det } \mathcal{H}_c = +1$ in every class. The example uses no transport solve and no training, and the signature is a property of the permutation connection alone.

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Data Availability. The synthetic dataset, the trained ensemble, and the numerical outputs behind every figure are generated by the source code deposited at [24] and are reproducible from it.

Code Availability. The numerical experiments reported in the manuscript are reproducible from the source code deposited at [24]. The code is sufficient to regenerate every figure and numerical value reported in the text.

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